Provisioning Recall: Windowed Decay-Attention Saturates Its Window, and How Much It Will Recall Can Be Predicted Before Any Training Result Exists

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Abstract

We study associative recall in small sequence models under a sealed, pre-registered protocol with hidden evaluation salts, matched parameter counts, and matched measured-FLOP budgets. We report four findings — one of them a supporting observation stated with its bounds — and one instructive failure.

(1) Saturation, measured by an instrument that controls retrieval distance — and a provisioning result pre-registered before any training result existed. A decay-attention layer whose causal read is restricted to a window of width W, trained past its compute threshold, attains recall equal to the fraction f of query-to-key retrieval distances the window covers, plus chance on the remainder: f + (1-f)·p_chance. Neither the functional form nor the underlying phenomenon is novel, and we say so before presenting evidence: the form is published [2606.02680], and sliding-window models failing to retrieve beyond their window is reported independently at least four times from 2024 onward [2401.01325; 2406.07522; 2512.10411; 2404.08865]. Our contribution is methodological: an instrument that holds retrieval distance fixed independently of context length — which existing long-context evaluations do not do; RULER concedes a "lack of position controlling" in its own limitations — and the pre-registered predictive use of the relation. The empirical claim is saturation: trained windowed layers land *on* the arithmetic ceiling rather than below it, extracting everything the window admits and nothing beyond. The relation retrodicts four prior cells to within 0.005, and in a pre-registered out-of-sample campaign — numeric ceilings for eight never-run cells across three task geometries, sealed and committed to a public ledger before any training result existed (externally timestamped from campaign ap onward; earlier campaigns are internally attested, §Reproducibility) — all eight matched within 0.005 against a ±0.05 tolerance (final aggregator decision, 48 runs). On natural text at 110M parameters, with the retrieval format supervised explicitly, all four windowed cells calibrate inside 0.005 and recall is a step function at the window — 1.000 inside, chance outside, with the boundary inclusive ($d \leq W$) in all three testable cells for our implementation; the pre-registered multi-seed replication has since reported (p1e, verdict LAW_REPLICATES_ACROSS_SEEDS): two further seeds across the same four windows pass all seven gates, every new cell calibrates inside 0.006, and the per-window spread across the three seeds is at most 0.0054 against a 0.10 gate — the natural-text leg is a property of the architecture-and-window, not of one model. On deployed checkpoints (Phi-3-mini, W=2047; Mistral-7B-Instruct-v0.1, W=4096) the same instrument finds a sharp recall cliff exactly where the architecture enforces a sliding window

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and none where it does not, but a small far-field residual is real and the strict pre-registered transfer gates fail in both arms: the paper accordingly does not claim the law transfers to deployed checkpoints. A conversion pilot then uses the law prospectively on a *converted* checkpoint — not zero-shot transfer: a deployed Qwen3-0.6B with all 28 layers rebuilt on the windowed module and retrained under a fixed two-stage recipe with the retrieval format supervised in the recovery corpus, its predicted aggregate recall committed to the public ledger before conversion training began and the observation landing 0.0031 away at W=1024 (24/24 strata; the co-registered W=512 cell 0.0061 away, 23/24), while delivering 1.75× and 2.44× the recall per byte of KV state of its two folklore comparator layouts at 12.5% of the base cache at 8K context. We state the certification's bounds with it: the window was declared rather than law-chosen, one training seed ran, and the conversion failed its pre-registered quality kill-gate — +0.343 nats of held-out LM loss against a 0.10-nat budget — so what is certified is the prediction, not a shippable model.

(2) The ceiling is conditional, and the condition is compute. Saturation describes where a *transitioning* run lands, not whether a run transitions. On the hard panels per-run outcomes are bimodal — near-ceiling or at chance, rarely between — so provisioning requires both a window wide enough and a budget above threshold. We publish the per-cell transition counts behind every ceiling we report (3, 6, 3, 6, 5, 6, 6, 6 of 6), because a maximum-over-seeds statistic alone would misrepresent this.

(3) A component perplexity misprices. Whether a model transitions at all is determined by local KV preprocessing — a short causal convolution, or the Forgetting Transformer's data-dependent KV-shift. Every configuration lacking it is 0/6 across our component matrix; the published base FoX formulation is 0/22 while FoX's published Pro assembly is 6/6; and in nine sealed campaigns, mixers carrying the full component set crossed in 133 of 144 runs against 1 of 144 matched gated-linear-attention controls. In FoX's own ablation the same component is worth 0.18 perplexity points — 6.62 to 6.80, about 0.027 nats — one contributor among several. A component whose removal moves the field's default metric by under three percent relative is a precondition under a recall endpoint. The decay gate's parameterization, by contrast, is immaterial: unbounded log-sigmoid, one-sided, and two-sided bounded forms each transition 6/6 inside the same shell.

(4) Where softmax attention's threshold lies — a supporting observation, stated with its bounds. On the hard multi-query associative recall (MQAR) panels a matched softmax-attention transformer crossed in none of 182 runs — 170 distinct seeds across 27 panel × budget × learning-rate cells, spanning measured budgets 2.8×10^{15}–1.32×10^{17} FLOPs and base learning rates 10^{-6}–10^{-2} across the 138 runs for which released artifacts record the rate (§4.10) — where the candidate crosses at 3×10^{15}: a tested separation of at least sixteenfold across the full learning-rate sweep (budgets to 4.8×10^{16}) and at least fortyfold at the step-matched learning rate (1.32×10^{17}, campaign az, 59,943 optimizer steps, spanning the transformer's observed transition window). This is a threshold claim, not an incapacity claim: on a deliberately easy panel the same transformer crosses in 2 of 6 seeds at 3×10^{16} FLOPs (recall 1.000, a textbook grokking event), under a pre-registered escalation ladder we ran against our own null. A pre-registered interior probe locates the hard-panel failure as representational — the answer token is not linearly decodable from any layer of a failed transformer (0.067–0.073 against chance 0.0625) — while the identical probe reads 0.977–1.000 in seed-matched candidates and 0.71–0.82 in transformers that did cross.

(5) A failure we report rather than bury. We attempted to locate the window boundary on the synthetic panels and could not. On a fixed-distance panel, a window reaching 157 tokens solved a 160-token retrieval 6/6, while a window reaching 163 — covering the target outright — failed 0/6. Window width governs optimization difficulty as well as reach, the two are confounded in that design, and the effect is documented elsewhere (SWAX, 2509.24552). The boundary question was later settled for our implementation by the inverted design on natural text, and the synthetic question is stated as open with the design that would settle it.

All results are at ~14M parameters on synthetic MQAR-family tasks except where stated. Campaigns r, s, t, and u were designed and executed fully autonomously by the resident LLM operator; the remaining campaigns were designed by a supervised interactive session and executed under the same sealed protocol — pre-registration, hidden evaluation salts, source hashing, external ledger timestamping, adversarial self-audit — which refuted three of our own conclusions during this work, including two that had reached a draft patent specification. We release the audit trail, both analysis scripts named in the Reproducibility section, and the full changelog.

1 Introduction

Sizing an attention window is a provisioning decision. The window determines which retrieval distances a model can serve, and current practice sizes it by convention or by sweeping trained models after the fact. The central result of this paper is that, for the models we study, the attainable recall of a windowed layer can be written down in advance: a decay-attention layer trained past its compute threshold saturates its window, landing on the ceiling f + (1-f)·p_chance, where f is the fraction of query→key retrieval distances the window covers — and this ceiling passed a pre-registered out-of-sample test in which every numeric prediction was sealed before any training result existed. The measurement behind that test is an instrument that holds retrieval distance fixed independently of context length, which, to our knowledge, no published long-context evaluation does.

Associative recall — storing key–value bindings in context and retrieving them on demand — is widely treated as the capability axis that separates softmax attention from its subquadratic challengers. The MQAR benchmark family [Zoology; Based] was constructed to measure this separation, and its published results consistently place attention at or near the ceiling: linear-attention and SSM variants are evaluated by how much of the attention–recall gap they close, as a function of model width, state size, and key–value load.

This paper reports four findings that complicate that picture. They are ordered by the defensibility of their contribution rather than by drama: the provisioning result and its instrument lead, and the compute-gated phase transition — an earlier draft's headline — is presented as the condition under which the ceiling applies, because abrupt recall acquisition along the training axis is by now well documented [2209.11895; 2507.01414; 2511.12768; 2511.16893] and our addition to it is the conjunction: per-seed bimodality, matched measured-FLOP budgets, cross-mixer comparison, sealed protocol.

Finding 1 (a provisioning instrument, and what is and is not new about it). For windowed reads the transition does not disappear but saturates at a ceiling recall = f + (1-f)·p_chance, with f the fraction of query→key retrieval distances covered by the window (§4.1). **We claim neither the functional form nor the underlying phenomenon as novel, and say so before presenting the evidence.** The coverage-plus-chance form is published: Yang [2606.02680] gives Acc ≈ π + (1-π)·a_uncovered with the K-way bound π + (1-π)/K, which is the same arithmetic, derived there from mask geometry for block-sparse attention. That sliding-window models fail to retrieve beyond their window is also established — Self-Extend [2401.01325] reports that Mistral with SWA "can only access information within its sliding window", Samba [2406.07522] that a fine-tuned SWA model "simply cannot recall memories beyond 4K length", and SWAA [2512.10411] that pure SWA scores 0.0 on RULER MQ-NIAH while a 1:1 global interleave scores 59.2. What we contribute is narrower and methodological: an instrument that holds retrieval **distance** fixed and independent of context length, which the existing evaluations do not — RULER's own limitations section concedes a "lack of position

controlling", and needle-depth heatmaps confound distance with length because the query sits at the end. Under that instrument the ceiling is *predicted before training rather than fitted after it*: it retrodicts four prior cells to within 0.005 and passed a pre-registered eight-cell out-of-sample test across three unseen task geometries, all errors $\leq$0.005 against a $\pm$0.05 tolerance. The contribution is the pre-registered predictive use and the distance-controlled measurement, not the equation. A conversion pilot (§4.3, campaign p2g) carries the predictive use onto a *converted* checkpoint: Qwen3-0.6B rebuilt on the windowed module and retrained under a fixed recipe with retrieval supervision in the recovery corpus, with predicted aggregate recall at two windows sealed before conversion training began and the observations landing 0.0031 and 0.0061 away — beside a quality kill-gate the conversion failed (+0.343 nats against a 0.10 budget), which bounds what that certification licenses (limitations 14 and 15).

Finding 2 (phenomenon: the compute condition). Recall acquisition at fixed model size is not a graded quantity but a *phase transition in training compute*. Across 200+ budget-halted training runs at matched parameters, per-run hidden-eval recall on the hard panels is bimodal: $\approx$1.0 or $\approx$chance ($1/16 = 0.0625$), with only 4 candidate runs caught between (all identifiable as mid-transition at budget exhaustion from their loss trajectories). On the hard panels, difficulty and budget modulate the *probability* of transition, not the attained recall level; on the easy panel a stable partial-retrieval regime exists and the attained level is itself budget-dependent (§4.4). Mean-based comparisons therefore conflate transition probability with effect size, and standard mean-gap statistics mischaracterize the regime; we develop transition-count statistics instead. This bimodality is the condition attached to Finding 1: the ceiling describes where a transitioning run lands, not whether a run transitions, so provisioning requires both a window wide enough and a budget above the threshold (§4.4).

Finding 3 (mechanism). A 17-campaign component matrix resolves the mechanism into a three-way dissociation (§4.6, §4.7). Neither softmax attention alone (transformer, 0/182 on the hard panels at budgets to 1.32×10^{17} FLOPs (to 4.8×10^{16} across the full learning-rate sweep) — though on an easy panel at 3×10^{16} it crosses in 2/6 seeds, so this is a threshold statement and not a categorical one) nor input-dependent decay alone (GLA, 1/96) crosses the gate; the combination of a *softmax read*, *input-dependent decay bias*, and *local preprocessing of the KV path* (short depthwise convolution, or FoX-Pro's data-dependent KV-shift) does, robustly. The specific decay parameterization is immaterial — bounded two-sided, unbounded log-sigmoid (FoX's), and one-sided gates all transition 6/6 inside the same shell — while removing preprocessing yields 0/6 in every configuration tested, and QK-/output-normalization modulate the transition rate. This explains a previously unexplained empirical fact: the published base FoX formulation fails here (0/22), yet FoX's own published Pro assembly succeeds (6/6). FoX's own Table 3 ablates these components on perplexity (KV-shift removal costs 0.18 perplexity points, 6.62 $\rightarrow$ 6.80, $\approx$0.027 nats) without a recall endpoint or a mechanistic account; what we add is that under a recall-transition endpoint the same component is not a marginal contributor but a precondition (0/6 in every arm lacking it).

Finding 4 (inversion — a supporting observation, stated with its bounds). The compute threshold at which the transition becomes reachable is architecture-dependent, and we now bracket it on both sides for one panel. On an easy MQAR panel the candidate mixer crosses at or below 3×10^{15} FLOPs (8/8 seeds in aw's stage 1, at 3×10^{15} itself; ax's seed-matched candidate arm additionally crossed 6/6 at the 3×10^{16} rung) while a matched transformer crosses in 0 of 8 seeds at 3×10^{15} and 2 of 6 seeds at 3×10^{16} — a measured separation of at least tenfold with both ends identified. On the harder `pairs80/p80x384` panels the candidate crosses at

3×10^{15} while the transformer does not cross in 182 runs at budgets to 1.32×10^{17} (to 4.8×10^{16} across the full learning-rate sweep; the top budget is campaign az's single step-matched rate) and learning rates spanning 10^{-6}–10^{-2}, placing the separation there at no less than fortyfold with the transformer's threshold still above the tested range. Gated linear attention essentially never crosses (1/96). We make no incapacity claim for attention: the harness produces the attention retrieval circuit when given enough compute, and the pre-registered falsification test that establishes this was run, and is reported here, before publication. To our knowledge, no published MQAR result shows any architecture acquiring recall at a compute budget where matched softmax attention has not.

A further contribution is methodological. Campaigns r, s, t, and u were designed, launched, monitored, and statistically gated end-to-end by an autonomous LLM operator ("archlab") with no human in the loop; the remaining campaigns were designed by a supervised interactive session and executed under the same sealed protocol: candidate mixers are proposed against a frozen API contract; claims require pre-registered transition-count gates, fresh never-reused seeds, hidden evaluation salts sealed before launch, and complete-pair accounting with no efficacy stopping; every decision, seed ledger, source hash, and gate is on disk in an append-only audit trail. The candidate architecture itself illustrates why this discipline matters: our adversarial related-work audit — run before any public claim — found that the LLM-proposed mixer independently converged on the central FoX idea [arXiv 2503.02130] of cumulative, data-dependent forgetting inside an attention-shaped read, and the subsequent attribution campaigns (§4.7) replaced an initially wrong attribution story with the dissociation above. The phenomenon, the inversion, the mechanism, the law, and the protocol are the contributions.

2 Related work

Sliding-window retrieval, and the coverage relation. That a pure sliding-window model cannot retrieve beyond its window is established and we do not claim it. Self-Extend [arXiv 2401.01325] states that Mistral with SWA "can only access information within its sliding window"; Samba [arXiv 2406.07522] that a fine-tuned SWA model "simply cannot recall memories beyond 4K length", and that perplexity hides the failure; SWAA [arXiv 2512.10411] tabulates pure SWA at 0.0 on RULER MQ-NIAH against 59.2 for a 1:1 global interleave; and a direct Mistral v0.1-vs-v0.2 comparison [arXiv 2404.08865] reports 40.9% versus 94.8% recall. That interleaved global layers restore retrieval is likewise reported [2512.10411; 2606.15378; 2501.18795; 2403.19887]. The theoretical receptive-field bound for stacked windows (reach $\leq$ layers $\times$ W) is from Longformer [arXiv 2004.05150] and is stated for LLMs in Mistral [arXiv 2310.06825]; that the bound is not achieved in practice is treated in [arXiv 2503.03588] and [arXiv 2406.14909]. **The coverage-plus-chance relation itself is published**: Yang [arXiv 2606.02680] gives Acc $\approx \pi + (1\text{-}\pi)\cdot$a_uncovered with the K-way bound $\pi + (1\text{-}\pi)/K$, derived from mask geometry for block-sparse attention. What this literature does not do — and what §4.1 turns on — is control retrieval *distance* independently of context length. RULER [arXiv 2404.06654] concedes a "lack of position controlling" in its own limitations, and needle-depth heatmaps confound the two because the query sits at the end, so distance $\approx$ (1 - depth) $\times$ length. Lost in the Middle [arXiv 2307.03172] documents the complementary positional effect — accuracy varies with where the answer sits inside contexts of fixed length, a U-shaped bias — but its control is the mirror of ours: it varies answer position at fixed context length, whereas our instrument holds retrieval distance fixed while context length varies; related but distinct controls, and neither substitutes for the other. Controlled-evaluation work is now converging on the same confound

from other directions: CRE [arXiv 2605.23170] audits eleven long-context benchmarks, reports that none jointly controls task position, filler content and context length, and builds a controlled framework — for reasoning tasks, over categorical position bins (beginning/middle/end); [arXiv 2510.05381] separates context length from retrieval difficulty by masking and shows length itself degrades performance even when retrieval is perfect. Neither holds needle-to-query distance fixed as a continuous variable on retrieval probes while total length varies independently, which is the specific control this paper's instrument provides and its provisioning use requires. Our contribution is that instrument and the pre-registered predictive use of the relation, not the relation.

Recall benchmarks and the attention ceiling. Zoology [arXiv 2312.04927] and Based [arXiv 2402.18668] establish MQAR as the discriminating task between attention and subquadratic mixers and document sharp performance walls as functions of model dimension, recurrent state size, and key–value pair count. Subsequent theoretical work proves capacity-axis phase transitions for recall [arXiv 2605.05189; 2605.10795]. Throughout this literature, softmax attention is the reference ceiling — Zoology reports attention solving MQAR at a constant model dimension of 64 across sequence lengths, which is the strongest single point of tension with our §4.10 result and which we address there directly. The *capacity*-axis papers above do not study the training-compute axis; work that does study recall emergence during training [arXiv 2507.01414, which times the recall phase transition per model size and finds larger models transition earlier; arXiv 2511.12768 on phase transitions in small transformers; arXiv 2511.16893 on predicting induction-head emergence] does so within a single architecture class rather than as a matched-budget comparison across mixers, which is the axis we study.

Grokking and emergent circuits. The induction-head phase change [Olsson et al., arXiv 2209.11895] and the grokking literature document abrupt capability acquisition during training. Olsson et al. also supply the closest published antecedent of this paper's KV-preprocessing finding: their "smeared key" architecture perturbation — mixing previous-token information into attention keys — moves the induction-head phase change earlier in training, with in-context learning moving in lockstep. That is the same class of lever §4.6–§4.7 isolate — theirs on the key path only, ours on keys and values — a one-token KV-shift or short convolution on the KV path; what we add is a sealed, matched-FLOP quantification of it at a recall endpoint, where the component moves from shifting a phase change to gating one (0/6 in every arm lacking it). Per-seed grok/no-grok bimodality has been reported [arXiv 2607.05104], including sensitivity to numerics. Adjacent studies dissect what controls whether and when such transitions occur: weight-decay-regime diagnostics [2605.20441], a systematic sweep attributing grokking onset primarily to optimization stability and regularization rather than architecture [2603.25009], and developmental trajectories showing capability- and circuit-emergence timing differs across 1B-class architectures [2606.02378]. Okpekpe & Orvieto [arXiv 2508.19029] caution that apparent architectural failures on recall tasks are frequently optimization artifacts — a critique we address directly with learning-rate sweeps for both baselines (§4.8). Their specific finding cuts in our favour and we rely on it: they report that *the learning rate is critical for modern recurrent models but that standard transformers are robust to it*, with recurrent success confined to a narrow learning-rate window. A transformer whose recall is flat across a 33× learning-rate range, as ours is, is therefore the behaviour their result predicts rather than a sign of an untuned baseline. They also report that attention on AR shows an induction-head-like bump in loss that does not translate into an accuracy improvement — closely resembling the flat-loss, chance-accuracy transformer runs we report in §4.10. The conjunction we report — bimodal per-seed transitions,

on the compute axis, compared across architectures under sealed matched-FLOP protocol — is to our knowledge unreported.

Decay and selectivity mechanisms. Data-dependent forgetting is the through-line of the modern subquadratic literature: GLA [Yang et al.], DeltaNet and Gated DeltaNet, Mamba's selectivity, HGRN's lower-bounded forget gates, Griffin's RG-LRU with bounded input-dependent gating, RetNet's per-head decay spectrum, and ALiBi's log-spaced slopes as a static analog. The Forgetting Transformer [FoX, arXiv 2503.02130] injects a data-dependent cumulative log-forget bias directly into softmax attention logits — the same functional form as our candidate's read — and its *Pro* variant adds QK-normalization, output normalization, a sigmoid output gate, and a data-dependent KV-shift. Our candidate combines a FoX-class read with a log-spaced static half-life prior (cf. ALiBi/RetNet), bounded multiplicative modulation (cf. RG-LRU), a short depthwise convolution, and a write gate. §4.6 and §4.7 show by a component matrix that the decisive ingredients in this compute regime are the softmax read, *some* input-dependent decay (FoX's unbounded gate, a one-sided Griffin/HGRN-style gate, and our bounded two-sided form are interchangeable), and local KV preprocessing (our convolution or Pro's KV-shift) — with the normalizations acting as transition-rate boosters. That K/V shifting is beneficial is itself established: [arXiv 2411.19574] ablates K and V shifts in standard attention and argues a shift lets a single layer implement what otherwise requires a two-layer induction circuit, predating FoX. What is new here is not that KV-shift helps but its role as the *necessary* component for the recall transition inside decay-attention assemblies — dissociated, by the swap and shell matrices, from the gate form (immaterial) and the normalizations (rate boosters). This locates the effect in the FoX-Pro family as a whole rather than in any single proposal's parameterization, and explains the base-FoX/Pro performance split that the FoX paper reports but does not mechanistically attribute.

Automated architecture discovery. ASI-ARCH [arXiv 2507.18074] demonstrated large-scale autonomous linear-attention discovery. Our contribution is not the automation of proposal but the automation of *disciplined confirmation*: sealed salts, pre-registered gates, fresh-seed ledgers, adversarial self-audits (learning-rate sweeps, related-work checks, refusal of numerically unsafe variants), and an append-only decision log.

3 Methods and protocol

3.1 Harness and constraint system

All experiments except the natural-text and deployed-checkpoint campaigns of §4.2–§4.3 run in archlab's S0 tier: pure-PyTorch models (d=384, 8 blocks, 6 heads, SwiGLU MLP, RMSNorm; ~14.3M parameters — 14,327,216 for the GLA arm on the main p80x384 panel, 14,346,720 for the candidate), no mixed precision, single-GPU training per run on NVIDIA V100-32GB. The harness enforces seven constraints: (C1) parameter counts matched across architectures to <2% by construction (shared embedding, block layout, and MLP; only the token mixer differs, with per-name deterministic re-initialization so shape-compatible tensors are bit-identical across paired arms); (C2) training halts at a fixed *measured*-FLOP budget with standardized accounting, so quadratic-cost mixers pay their true cost; (C3) data are generated deterministically from (salt, seed, step) — the corpus is frozen by construction; (C4) hidden evaluation batches are generated under a salt sealed in a file the training stream never reads, fresh per campaign; (C5) multi-seed protocols with paired common initialization; (C6) automatic bit-level reproduction checks; (C7) an efficiency-adjusted composite score for the discovery loop.

3.2 Tasks

The frozen suite comprises MQAR (key–value pairs presented once, queried after a separator; loss on answer positions; chance = 1/16), a length-generalization variant, a state-tracking task (saturated at 1.0 by all architectures studied here, and thus not further discussed), and a permuted continual-learning phase with retention scoring. Panels are parameterized by sequence length, pair count, query count, and key-vocabulary size; the panels used here are p44 (160 tokens/44 pairs/64 keys), pairs80 (320/80/128), seq384 (384/64/128), p80x384 (384/80/160), and pairs48x448 (448/48; used only in the out-of-sample law test, §4.1).

3.3 Architectures

The registry provides transformer (softmax attention with RoPE, layout AAAAAAAA), GLA (data-dependent scalar-per-head decay — the coarse per-head gate of the Mamba-2/Gated-DeltaNet style rather than GLA's per-dimension gate [2312.06635], and without DeltaNet's delta-rule update — short (k=4) depthwise convolution, per-head output norm; GGGGGGGG), a 3:1 hybrid (GGGAGGGA), and candidate mixers loaded against a frozen contract (CCCCCCCC). The primary candidate, StableGLA, computes qkv from a short depthwise convolution; forms logits log softplus($q \cdot k^T/\sqrt{d}$) + (Λ_i - Λ_j) + log σ(write-gate), where Λ is the cumulative sum of per-head negative decay rates given by a static log-spaced half-life prior (8→256 tokens) modulated multiplicatively by exp(0.5·tanh($W_a x$)); and reads with row-softmax over the causal mask. Its read is $O(T^2)$ and FLOP-charged as such (C2). The component matrix of §4.6/§4.7 additionally evaluates: a *minimal architecture* (short convolution + input-dependent decay bias + softmax read + QK-norm + output-norm; no static prior, no softplus kernel, no write gate); shell variants swapping the decay gate's form (FoX's unbounded log-sigmoid; one-sided $\sigma(u)$); a faithful base-FoX baseline (cumulative log-sigmoid forget bias on standard scaled-dot-product logits; no convolution, no prior, no write gate); and a FoX-Pro-exact baseline per the published Pro block (QK-norm, output norm, sigmoid output gate, data-dependent KV-shift, unbounded gate). Windowed variants restrict the softmax read to a causal window W.

3.4 Sealed transition-count protocol

Because outcomes are bimodal (§4.4), claim campaigns pre-register: a transition threshold (hidden-eval recall > 0.8); minimum candidate transitions and maximum control transitions out of N fresh paired seeds; a one-sided Fisher exact bound; a worst-pair floor (Δrecall $\geq$ -0.10); and complete-pair accounting. Gates are written into the campaign manifest before launch, with fresh salts and seeds drawn from a global ledger — unique within a campaign, and deliberately shared between a campaign's two paired arms, since pairing on seed is the design. An audit on 2026-07-31 found two cross-campaign collisions (seeds 830–835 in campaigns o and p; 1400–1405 in campaigns aq and ar). Both fall on different arms of different campaigns, so no single comparison double-counts a seed; they are disclosed rather than removed, and earlier drafts' claim of "never-reused seeds" was incorrect; amendments are logged with timestamps and scientific-impact statements; there is no efficacy stopping. Because both arms run on the same seed set, the design is matched-pairs and the primary statistic is the exact McNemar test (a binomial sign test on discordant pairs). Earlier drafts made Fisher's exact test primary and described ignoring the pairing as conservative; that is wrong in the realized direction. Fisher's null permutes outcomes across all 32 runs, presuming runs from different seeds are exchangeable, which the design does not license; it returns p-values about four orders of magnitude smaller than

the paired test. Fisher values are retained in §4.5 only as a labelled unpaired sensitivity analysis. This changes no conclusion — every sealed campaign remains decisive — but it changes the reported strength of evidence. Non-claim campaigns (calibration, factorization, ablation, sweeps) are marked claim-ineligible in their manifests. The out-of-sample law test (§4.1) additionally pre-registers *numeric predictions* per cell, sealed in the campaign manifest and committed to the append-only ledger repository before training. For the record, the manifests' pre-registered primary was the one-sided Fisher exact bound at $\alpha = 0.005$; the exact McNemar test's primacy is this paper's analysis-level correction, not a pre-registered choice, and §4.5 states this beside the table it governs.

3.5 Family-wise error across the claim-eligible tests

This paper reports roughly fifteen claim-eligible pre-registered tests: the ten sealed confirmation campaigns (the nine GLA-controlled campaigns of §4.5's table plus the FoX-Pro-versus-transformer campaign aq, §4.10), the interior probe av, the falsification-ladder stages aw and ax, the step-matched control az, and the out-of-sample law test ap (the probe calibration ay accompanies av rather than gating a separate claim). One flag deserves a note so the artifact record is not misread: ap's manifest sets `claim_eligible: false`, because that flag gates the candidate-versus-control architecture-claim machinery of §3.4, which ap does not use; ap is a sealed prediction test with its own pre-registered numeric gates, and it is counted in this statistical family on that basis. Each was gated at its own pre-registered α, which is the design under which the campaigns were sealed; we did not pre-register a family-wise correction, and we disclose that rather than apply one silently after the fact. A reader who demands one can apply Bonferroni $\times 15$ without changing any headline result: the sealed-series McNemar values (9.8×10^{-4} and below) and ap's eight-cell calibration survive it trivially. Two results are marginal under that correction and we name them: campaign av's $p = 3.9\times10^{-3}$ and campaign aw's Stage-1 $p = 3.9\times10^{-3}$ each correct to ≈ 0.059, just above 0.05. av's mechanism reading does not rest on that p-value alone — it is calibrated within-architecture by campaign ay (§4.10) — and aw Stage 1's role in the ladder was to trigger Stage 2, not to carry a claim. Campaign ax's Stage-2 $p = 0.0625$ is non-significant before any correction and is already reported as such (§4.10). No conclusion in this paper changes under the $\times 15$ correction; the reader should nonetheless know which results sit near the boundary.

3.6 The autonomous protocol

The full campaign graph — calibration (k), cliff factorization (l), sealed confirmation (m), attention controls (n), ablations (o), hardening (p), the q/r/s/t/u sealed series, both learning-rate sweeps, the attribution and component-matrix campaigns (y, z, aa–af, an, ao), the minimal and windowed seals (ag, aj, ak, am), and the law test (ap) — was executed over roughly four days on four V100s. **Autonomy scoping, stated precisely:** campaigns r, s, t, and u were designed, launched, gated, and reported end-to-end by the resident LLM operator with no human in the loop, following written heuristics and hard limits; all other campaigns were designed by a supervised interactive session (human plus LLM) and executed under the same sealed pre-registration protocol and audit trail. We make no "all experiments were autonomous" claim. The operator's decision log records, among other things, three consecutive refusals to implement a chunked-linear kernel on documented correctness grounds (the softplus content term is not factorizable), seed-ledger and source-hash verification before every launch, and

immediate notification triggers for claim-affecting outcomes. We regard the earlier silent failure of a manually-launched diagnostic (32 workers dead on a missing dependency while logs read "Completed") — and its detection by audit — as part of the case for this discipline. So is a second, subtler instance. The retrofit pilot in the adjacent lane reported zero key–value accuracy after 900 training steps with 30% key–value-formatted exposure. The cause was a label mask that zeroed precisely the position whose target was the answer token: the model was taught the prompt template while no gradient ever reached the key→value binding, making improvement structurally impossible. The fix was two lines, and it was verified concretely rather than argued — real checkpoint, real generated examples, loss 4.67 → 0.25 with exact-match recovery — after which every pre-fix number in that pilot was voided rather than reported. Two of the three errors this protocol has caught in itself were **false nulls**, which is the failure mode a programme built on negative results should fear most, and is why auditing an instrument on a trained checkpoint before trusting a null is now a standing rule. So is the correction this version embodies: v0.3's FoX attribution claim was refuted within a day by the program's own adversarial campaigns (an/ao) before any public claim was made. The complete audit trail (manifests, salts, seed ledger, decision JSONs, per-run training logs) is released with this paper, except where the Reproducibility section states an item is retained and available on request.

4 Results

Results are ordered with the provisioning law and its instrument first (§4.1–§4.3), the compute condition and the mechanism next (§4.4–§4.8), and the boundary failure and the transformer's threshold — supporting results — last (§4.9–§4.12).

4.1 A predictive window-provisioning law

Restricting the minimal architecture's softmax read to a causal window W does not destroy the transition; it caps it. (The minimal architecture is defined in §4.6, and "transition" is the compute-gated event of §4.4–§4.5; both sections follow this one in the reframed order, and the law is stated first because it is the paper's lead result.) Transitioned windowed runs saturate not at 1.0 but at a plateau that is tight across seeds (e.g., 6/6 seeds within 0.758–0.761 on seq384/W128) and panel-dependent. The plateau is quantitatively predicted by a **window-provisioning law**:

> **recall ceiling = f + (1 - f)·p_uniform**, where f is the fraction of query→key retrieval distances ≤ W under the task generator, and p_uniform = 1/16 is the rate of a uniform guess over the value alphabet. In-window pairs are retrieved; out-of-window pairs fall to the uniform-guess rate. We write p_uniform rather than "chance" deliberately: on these panels the *Bayes-optimal* no-retrieval rate is strictly higher than 1/16, and the empirical content of the law is that trained models land on the lower of the two. §4.1.1 measures both and shows which one the models occupy.

Retrodictive agreement (four cells). Computed after the windowed campaigns had run, the law reproduces every observed plateau: seq384/W128 predicted 0.7583, observed 0.7581–0.7614; pairs80/W192 predicted 0.9119, observed transitioning seeds 0.9109–0.9114; pairs80/W128 predicted 0.5723, observed ≤0.571; and all W=256 cells (window ≥ maximum retrieval distance, f=1) at 1.0. This agreement is *retrodictive* — the cells existed before the law — and we label it as such; it also resolves an apparent anomaly, since W192 transition counts were ceiling-limited (threshold 0.8 < ceiling 0.912) and seeds transitioned *to the ceiling*.

Pre-registered out-of-sample test (campaign ap). To elevate the law from retrodiction to prediction, campaign ap pre-registered numeric ceiling predictions for eight cells never previously run — three task geometries (pairs80/seq320, seq384, and a new pairs48/seq448 panel) crossed with unseen windows W ∈ {64, 96, 128, 160} — computed from generator geometry alone before any training, sealed in the campaign manifest (registered 2026-07-30T02:55Z) and committed to the append-only ledger repository (timestamped commit) before results existed, with a ±0.05 per-cell tolerance, a trivial control (f=1 cell must reach ≥0.95), and a ≥6-of-8 pass gate. Because the law predicts a *ceiling* and individual seeds may fail to transition, the pre-registered per-cell statistic is the maximum over 6 seeds.

Cell	f (in-window fraction)	Predicted ceiling	Observed max (final)	Error
pairs80 / W96	0.344	0.3852	0.3872	+0.0020
pairs80 / W160	0.744	0.7602	0.7583	-0.0019
seq384 / W96	0.493	0.5243	0.5253	+0.0010
seq384 / W160	0.934	0.9384	0.9431	+0.0047
pairs48x448 / W64	0.406	0.4433	0.4456	+0.0023
pairs48x448 / W96	0.740	0.7563	0.7603	+0.0040
pairs48x448 / W128	0.969	0.9705	0.9724	+0.0019
pairs48x448 / W160 (control, f=1)	1.000	1.0000	1.0000	0.0000

Campaign ap's final aggregator decision (48 runs) is now complete and reproduces the interim maxima: all eight cells — including all three on the never-before-used pairs48x448 geometry — match their sealed predictions within 0.005, an order of magnitude inside the ±0.05 tolerance.

What the max-over-6 statistic does and does not say. Because the pre-registered per-cell statistic is the maximum over 6 seeds, it reports the ceiling of *transitioning* runs, not an expectation over runs. The number of seeds actually reaching their cell's ceiling (within ±0.05) was, in table order, 3, 6, 3, 6, 5, 6, 6, 6 of 6. The three cells with partial counts are exactly those whose predicted ceiling lies below the 0.8 transition threshold, and their non-transitioning seeds sit at or below the uniform-guess rate (0.062–0.078 on `pairs80`, 0.101 on `pairs48x448`), well beneath the no-retrieval floors of §4.1.1; one `seq384` seed at 0.132 sits at that panel's floor of 0.134 and is the corpus's only sub-threshold run consistent with counting rather than guessing — the same bimodality documented in §4.4, not intermediate values. The law therefore predicts *where a transitioning windowed run lands*; it does not predict *whether* a given run transitions, which remains governed by the compute threshold of §4.5. Provisioning use accordingly requires a budget above that threshold (§4.4).

4.1.1 The floor is not the chance rate, and the models do not find it.

The generator draws values i.i.d. with replacement from an alphabet of 16 (`data.py`: `vals = rng.integers(0, NVAL, size=pairs)`), so the values present in any one sequence form a *multiset*, not a set. A model that never matches a key can still do better than a uniform guess: it can read the multiset, eliminate the values already consumed by earlier answers, and

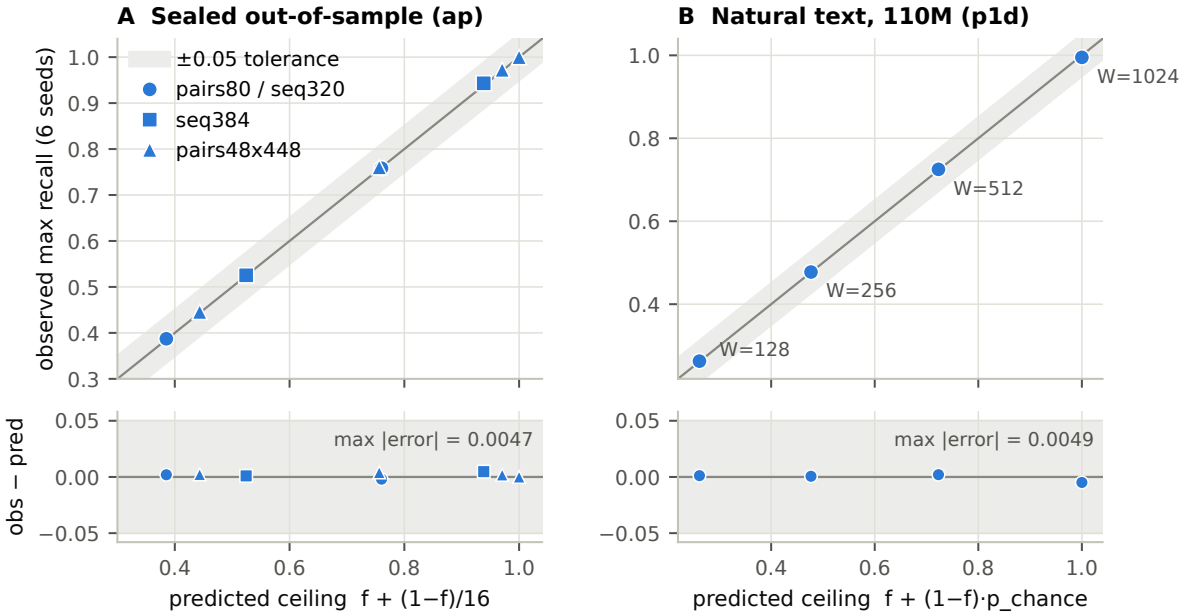


Figure 1: The window-provisioning law, predicted versus observed. **(A)** The eight sealed out-of-sample cells of campaign ap, across three task geometries: numeric ceilings $f + (1 - f)/16$ were committed to an append-only ledger before any training result existed, and the observed max-over-6-seeds recall matches every cell within 0.005 (residual strip, bottom) against a pre-registered ±0.05 tolerance. **(B)** The same law on natural text at 110M parameters (campaign p1d, seed 2100): all four windowed cells calibrate inside 0.005, at 8× the parameter count of the synthetic test.

emit the mode of the remainder. We had previously reasoned about no-retrieval performance from a distinctness assumption and stated a safety rule, `pairs - queries + 1` $\geq$ `NVAL`, that follows from it. **That rule is unsound and we withdraw it.** With replacement there is no distinctness to exploit, and every panel in this programme carries a no-retrieval floor above 1/16. We now compute the floor by simulation rather than argue it; the script is released in the ledger repository (§Reproducibility):

panel	pairs/queries	uniform guess	Bayes no-retrieval floor
easy48	8/8	0.0625	**0.4235**
pairs80	80/40	0.0625	0.1260
seq384	64/32	0.0625	0.1341
pairs48x448	48/24	0.0625	0.1461

This makes the eight-cell test of §4.1 sharper than it was designed to be, because the two hypotheses make different numeric predictions. Substituting the floor for 1/16 raises every predicted ceiling, and in five of the eight cells the two predictions separate by more than 0.016 — enough to discriminate against a per-cell error of $\leq$0.005. In all five the observed maxima land on the uniform-guess prediction and reject the floor prediction by 0.018–0.048: `pairs80`/W96 observed 0.3872 against 0.3852 (uniform) and 0.4269 (floor); `seq384`/W96 observed 0.5253 against

0.5243 and 0.5606; `pairs48x448`/W64 observed 0.4456 against 0.4433 and 0.4931; `pairs80`/W160 and `pairs48x448`/W96 likewise. The remaining three cells have f $\geq$ 0.93, where the predictions differ by under 0.005 and no discrimination is possible; we report them as uninformative on this question rather than counting them.

The pre-registered predictions were sealed with 1/16 and are unchanged. What has changed is that we now know what the alternative was, and that the data rejects it out of sample. **A trained windowed layer, denied its window, guesses uniformly — it does not exploit the in-context value statistics that are demonstrably available to it.** The saturation claim is therefore two-sided and both sides are measured: the layer extracts everything the window admits, and nothing beyond it, including statistics it could have counted for free. We did not design this test and record that it arrived by audit rather than by foresight.

Relation to coverage baselines, and to prior art we do not claim over. The functional form — covered fraction plus chance on the remainder — is the natural construction and appears in retrieval evaluation, where success rates are routinely compared against a random-selection baseline over the uncovered set. It has also been published in essentially this form for attention specifically: Yang [2606.02680] states Acc_D(M) $\approx$ π_D(M)·a_covered + (1-π_D(M))·a_uncovered with the K-way bound π + (1-π)/K, where π is the probability that a source–target pair is covered by the model's attention geometry. That is our f + (1-f)·p_chance. We became aware of it through an adversarial prior-art sweep run against our own claim, and we record it here rather than let a reader find it. Three things distinguish what we do from it, and none of them is the equation: that work derives π from mask geometry whereas we measure f from a task's retrieval-distance distribution; it uses the relation diagnostically and never to size a window in advance; and it studies block-sparse attention on GPT-2-scale models trained from scratch rather than native sliding-window attention. What is not standard, and what we test, is that a *trained* windowed attention layer lands on that arithmetic ceiling rather than below it: the claim is saturation, i.e. that the layer extracts everything the window admits and nothing beyond it, which is an empirical property of the trained model and not a property of the formula. The pre-registered test is of saturation, not of arithmetic.

The law makes window provisioning an *engineering calculation*: given a workload's retrieval-distance distribution, W can be chosen to meet a target recall, and the honest complexity statement for the windowed architecture is O(T·W) with W set to the required memory horizon.

4.2 The law on natural language, and the window boundary settled for this implementation

Everything above is synthetic and ~14M parameters. Campaign p1d tests the same law on a 110M-parameter windowed model trained on natural text, with the retrieval format supervised explicitly because a model trained on ordinary web text alone does not acquire it (§4.11; §5, limitation 1). Four cells, W $\in$ {128, 256, 512, 1024}, one window per cell, seed 2100, 1.2B tokens at context 1024, 73,243 steps, training recipe identical across cells. Scoring is evaluation-only against a frozen, salted battery whose predictions were fixed before any checkpoint was scored against it — p1c's coarse needle accuracies for these same checkpoints were already known when the battery was designed, and the battery is a redesign responding to p1c's instrument faults, so the genuinely falsifiable content here is the step-function exactness and the boundary battery rather than the coarse calibration levels: **Battery A**, 2,048 needle probes stratified over 32 retrieval distances from 24 to 1,000 tokens; **Battery B**, probes placed at and immediately around each window boundary; **Battery C**, a 128-probe negative control. The needle word pool gives a

chance rate of 1/65 = 0.0154, so the general statement of the law is

recall ceiling = f + (1 - f)·p_chance

of which §4.1's f + (1 - f)/16 is the 16-way special case. All six pre-registered gates pass.

The mechanism claim: recall is a step function at the window, not a decay. This is the result to read first, because the aggregate ceiling of the previous section is consistent with many mechanisms and this is not. Partitioning Battery A at each cell's own window:

Cell	accuracy, distances inside W	accuracy, distances outside W	negative control	probes in / out
W=128	1.0000	0.0177	0.0156	448 / 1472
W=256	1.0000	0.0165	0.0156	896 / 1088
W=512	1.0000	0.0226	0.0156	1408 / 576
W=1024	0.9951	— (no probe exceeds W)	0.0078	2048 / 0

The gate deliberately excludes a ±16-token band around each window (it compares d < W-16 against d > W+16), so the in/out counts do not sum to 2,048; the excluded band is reported separately below, where it is the interesting part rather than a nuisance.

Inside the window the model is at or within half a percent of perfect; outside it, it is at the negative control, which is chance. There is no intermediate regime and no graded decay with distance. The measured chance rate over the out-of-window probes, 0.0137, agrees with the assumed 0.0154 to within one probe's worth of resolution.

Aggregate calibration across an eightfold window range. The ceilings follow:

Cell	f	Predicted	Observed	Error
W=128	0.2500	0.261538	0.262695	+0.00116
W=256	0.4688	0.476923	0.477539	+0.00062
W=512	0.7188	0.723077	0.725098	+0.00202
W=1024	1.0000	1.000000	0.995117	-0.00488

Every cell is inside 0.005 against a pre-registered ±0.05 tolerance — the same order of agreement the synthetic out-of-sample test reached, now at 8× the parameter count on natural text. The adjacent gaps are 0.215, 0.248 and 0.270, all far above the pre-registered 0.15 ordering threshold. The only cell that undershoots is W=1024, by 0.0049, or ten probes of 2,048.

The window boundary is inclusive, and this settles a question §4.1's synthetic campaigns left open. Battery B is the design limitation 5 called for and campaign `ar` (§4.9) could not deliver: it holds the window fixed and sweeps the retrieval distance across the boundary, so window width no longer co-varies with optimization difficulty. Reading each cell on its own boundary:

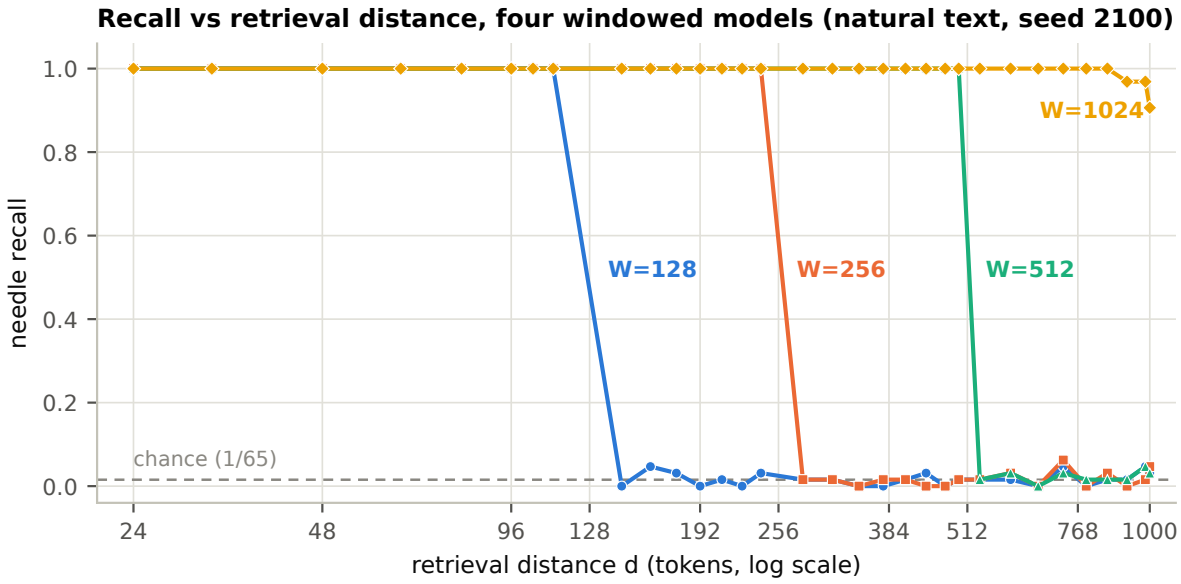


Figure 2: Recall is a step function at the window, not a decay. Per-stratum recall on campaign p1d's Battery A (2,048 salted needle probes stratified over 32 retrieval distances) for the four windowed 110M-parameter models: 1.000 at every stratum inside each model's window, the negative-control chance rate (1/65) outside it, with the step at each cell's own W. There is no intermediate regime and no graded decay with distance.

distance d relative to that cell's W	W-8	W-4	W-2	W-1	**W**	W+1	W+2	W+4	W+8
W=128 cell	1.00	1.00	1.00	1.00	**0.922**	0.016	0.000	0.031	0.000
W=256 cell	1.00	1.00	1.00	1.00	**1.000**	0.000	0.000	0.000	0.016
W=512 cell	1.00	0.984	0.984	1.00	**0.906**	0.031	0.016	0.000	0.016

Retrieval is intact at distance exactly W and collapses to chance at W+1, in all three cells. The correct convention is therefore $\mathbf{d \leq W}$, not the strict d < W the predictions assumed. Battery A cannot see this — none of its 32 strata sits exactly on a window — which is precisely why Battery B exists.

This also explains, rather than excuses, campaign p1b's one failed cell. p1b's W=512 gate failed by overshooting its prediction (0.594 observed against 0.415 predicted, §5 limitation 5), and its distance histogram placed 26 of 128 probes at a distance of exactly 512. Those 26 probes were scored as out-of-window by a strict-< prediction and are in fact in-window; crediting them moves the prediction to ≈0.62, which is close to what p1b measured. **p1b's recorded gate outcome**

is not amended — it failed and it stays failed — but the failure is now attributable to an off-by-one in the predictor rather than to the model exceeding its window.

We note what this is *not*. A reach beyond W would be the signature of a depth-relay receptive field, which for stacked sliding-window attention is bounded by window × depth and would extend recall well past the boundary. Recall here stops dead one token past W in every cell. The effect is an inclusive mask, not a receptive field, and earlier internal notes filing it as a receptive-field observation were wrong.

What this rests on — no longer one model. Everything above was measured on one model per cell, seed 2100 — the n=1 limitation the synthetic law does not have. Campaign p1e was the pre-registered replication: seeds 2101 and 2102 across the same four windows, training recipe unchanged, readings fixed before data, no efficacy stopping. It completed 2026-08-04T01:37Z with verdict **LAW_REPLICATES_ACROSS_SEEDS** — all seven pre-registered gates (G0–G6) pass. Before any new checkpoint was scored, the run re-scored seed 2100's checkpoints on the same battery and reproduced this section's four values exactly (agreement within 10^{-9}) — the pre-registered defence against the instrument silently reading its window from an import-time global — and the key–value positive control that p1b had failed reads 1.000 in all eight new cells rather than being assumed. The replicated ceilings:

Cell	Predicted	seed 2100	seed 2101	seed 2102	cross-seed spread
W=128	0.261538	0.262695	0.260254	0.257324	0.005371
W=256	0.476923	0.477539	0.477051	0.479980	0.002930
W=512	0.723077	0.725098	0.723145	0.721680	0.003418
W=1024	1.000000	0.995117	0.994629	0.994141	0.000977

The largest calibration error among the eight new cells is 0.005859 (seed 2102, W=1024), against the ±0.05 tolerance. The cross-seed gate — the one that earns the word "law", since it asks whether model-level variance is small compared with the between-window structure — required the per-window spread across the three seeds to stay within 0.10 against predicted between-window gaps of 0.21–0.27; the observed spreads, at most 0.005371, are roughly forty times smaller than the smallest between-window gap. Every adjacent gap in both new seeds exceeds 0.216 against the 0.15 ordering threshold; the step function reproduces (in-window accuracy ≥ 0.9941 in every new cell, out-of-window accuracy 0.010–0.022 against negative controls of 0.000–0.031); and the descriptive boundary battery is consistent with the inclusive convention in all six new testable cells, though not uniformly at ceiling: accuracy at distance exactly W is 1.000 in four cells and 0.8125 and 0.6875 in the other two (seed 2102 at W=128 and W=256) — all far above the $\leq$0.016 accuracy at W+1, where every cell collapses — including recall of 1.000 at distance exactly 512 in both new W=512 cells. The pre-registered reading, fixed before data, is quoted rather than paraphrased: "The window-provisioning result may be stated as a property of the architecture-and-window rather than of one model, and p1d's n=1 limitation is discharged rather than merely disclosed" (§5, limitation 13).

4.3 The instrument on deployed checkpoints: the dissociation holds, the transfer is not claimed, and a provisioned conversion lands on its sealed prediction

Everything above concerns models trained under our own harness. The p2 series points the instrument at released checkpoints, asking whether the machinery of §4.1–§4.2 — distance-

stratified probes, a structural window check, the coverage ceiling — behaves lawfully on models we did not train. Three campaigns ran as an instrument-repair chain (p2a → p2b → p2c), each fault documented in the successor's manifest and no gate ever amended in place; the chain is the protocol of §3.6 operating as intended, and its verdicts are reported exactly as the pre-registrations fixed them. A final campaign, p2g, inverts the question: rather than probing a released checkpoint as it stands, it converts one to the windowed module, retrains it, and asks whether the law could say in advance what the converted model would recall; it closes this section.

p2a: the four-way dissociation, qualitatively confirmed; its positive arms void. Four arms, evaluation-only, needle retrieval in natural filler text with retrieval distance recorded at construction rather than inferred: Phi-3-mini-4k-instruct (declared sliding window 2047, 32 layers) and Mistral-7B-Instruct-v0.1 (window 4096, 32 layers) as positive arms predicted in advance to cliff at their windows; Mistral-7B-Instruct-v0.2 (same lineage, window removed) and Gemma-2-2b-it (config declares sliding_window=4096, but local layers alternate with full global layers) as negative arms predicted not to cliff. The registered qualitative prediction confirmed on all four arms: a recall cliff appears exactly where the architecture enforces a window, and nowhere else — Mistral v0.2 recalls 1.000 at every distance tested including 6,144, and the Gemma-2 hybrid shows no cliff at its declared window. The controlled comparison carrying that result is v0.1 against v0.2: the same model lineage, differing in the window. p2a's quantitative gates were nonetheless not met, and its positive arms are void on the structural gate: the mask check had crashed on a misspelled keyword argument and silently fallen back to a behavioural probe that is invalid for stacked windowed layers, and distance was anchored to the needle's first token rather than to its value token.

p2b: the instrument repaired, the pre-registered transfer verdict negative. p2b re-ran the two positive arms with a verified structural gate — the model's own sliding-mask constructor is materialized and inspected, position i-W-1 blocked and i-1 visible, values recorded verbatim — and value-token distance anchoring. Its descriptive boundary battery resolves the cliff at one-token resolution and places it one to four tokens *before* W in value-token distance (Phi-3: 1.000 at d=2043 → 0.042 at d=2046; Mistral: 0.979 at d=4092 → 0.062 at d=4095): the needle spans ~10 tokens, and it is the key tokens — the associative anchor, earlier in the needle — that must sit inside the window. Its pre-registered quantitative gates were both negative: residual far-field recall exceeded the floor gate, and aggregate recall fell 0.082–0.096 below the strict-form calibration prediction. **p2b's pre-registered outcome is LAW_DOES_NOT_TRANSFER, and it stands unamended.** A subsequent audit then found two instrument faults in p2b itself: substring-match scoring that manufactured far-field hits ("789." credited as the value 78), and a negative control that was not length-matched, underestimating chance exactly where the far-floor gate needed it.

p2c: the corrected instrument (completed 2026-08-02T00:09Z). p2c re-ran both arms with exact-match scoring, per-stratum length-matched negative controls, and 48 repetitions per stratum across 24 strata — 1,152 scored probes plus 288 no-needle controls per arm. Its gates read, mechanically: **G0** (capability) and **G1** (structural window enforcement) pass in both arms, the materialized masks giving a maximum attendable distance of 2,046 (Phi-3) and 4,095 (Mistral). **G2a** (near-field plateau) passes in both arms: in-window recall is 1.000 at every stratum, except a single Phi-3 stratum at d=433 that reads 0.979 — one probe of 48. The cliff brackets the structural boundary exactly: Phi-3 holds 1.000 at d=1,999 and reads 0.062 at d=2,095, with the mask's last attendable distance (2,046) between them; Mistral holds

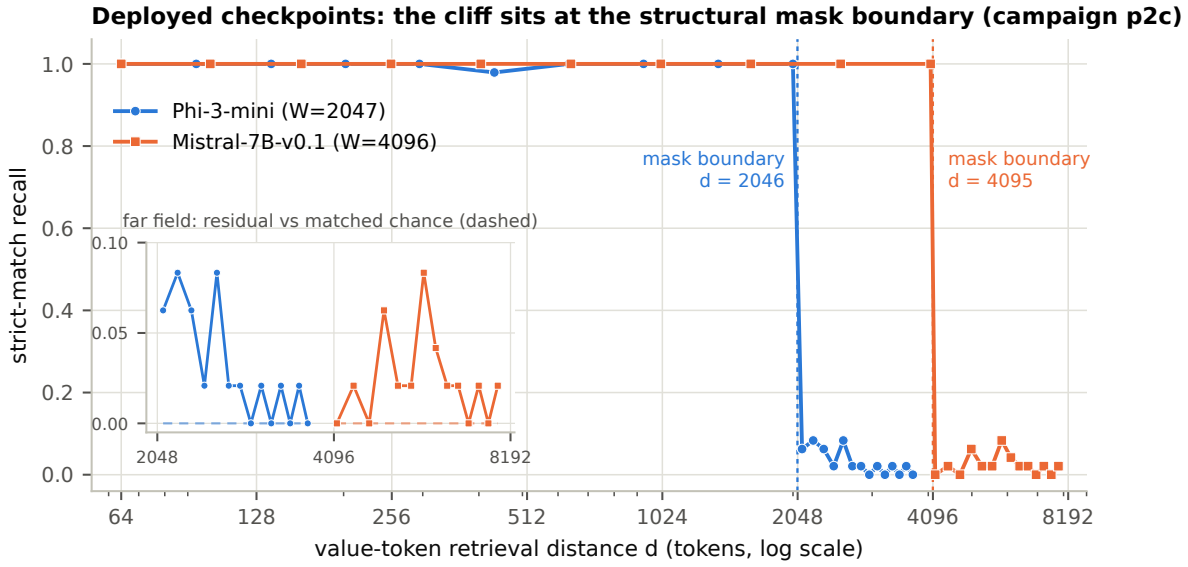


Figure 3: The instrument on deployed checkpoints (campaign p2c). Strict-scored recall versus value-token retrieval distance for Phi-3-mini ($W = 2047$) and Mistral-7B-Instruct-v0.1 ($W = 4096$), 48 exact-match probes per stratum: a sharp cliff brackets each architecture's structural mask boundary (dotted verticals at maximum attendable distance 2,046 and 4,095). Inset: the far field, where a small residual (mean ≈ 0.024–0.030, maximum 0.083) sits above per-stratum length-matched chance (dashed) — real on the corrected instrument, and the reason the strict pre-registered transfer gates fail.

1.000 at d=4,048 and reads 0.000 at d=4,144, against a structural boundary of 4,095. **The far-field residual survives the corrected instrument, and is therefore real rather than a scoring artifact:** mean recall over the far strata is ≈0.030 (Phi-3) and ≈0.024 (Mistral), maximum 0.083, against measured length-matched chance of 0.000 and 0.003. **G2b** (far field within 0.05 of measured chance at every stratum) accordingly fails in both arms — four strata violate it for Phi-3, two for Mistral. **G2c** (a sharp step to chance within a four-token stratum gap near the boundary) also fails in both arms, in part by construction, which we state rather than absorb: the runner as executed drew log-spaced strata with a ≈50-token guard band around the boundary instead of the per-integer boundary strata the manifest registered, so no stratum pair the step gate could accept existed, and the registered multi-anchor geometry sweep (G4) was not computed — the one-token boundary resolution above therefore rests on p2b's descriptive battery, not on p2c. The interpretation was fixed in advance: if G2c fails in both arms, p2b's outcome stands. **It stands: for the strict step-to-chance form, the law does not transfer to these deployed checkpoints, and this paper does not claim otherwise.**

What p2c does and does not license. Its calibration gate **G3** passes in both arms — aggregate recall within 0.017 of f + (1-f)·p_chance — but the boundary term in that prediction is *fitted*: b is estimated from the same data, as the largest stratum with recall ≥ 0.90. G3 is therefore instrument characterization, not a pre-registered prediction, and it earns no transfer claim. Its direction is still informative: both aggregates sit *above* the fitted prediction (+0.017 Phi-3, +0.014 Mistral) — on deployed checkpoints the law under-promises, because the residual adds recall beyond the ceiling and nothing is lost inside the window — so a provisioning decision

made with it errs conservative. What is currently claim-grade from the p2 series is narrower: the four-way dissociation (a cliff where a window is enforced, none where it is not), and the boundary-convention observation — the behavioural cliff sits at the *structural* mask boundary, which on these released checkpoints is exclusive at W-1, the opposite convention to our own implementation's inclusive $d \leq W$ (§4.2), confirming that the boundary convention is per-implementation engineering rather than a finding (§5, 2026-08-01 addendum). The test that could earn the transfer sentence is a pre-registered **p2d**: the instrument frozen exactly as p2c leaves it, numeric predictions fixed in advance from p2c's fitted boundary and residual term, fresh probe draws. Until such a campaign passes, every sentence in this paper about unconverted deployed checkpoints probed zero-shot (p2a–p2f) should be read as instrument characterization, and none as transfer. That campaign has since run; the next paragraph reports it under this document's dated-corrections convention.

p2d and p2e: the frozen predictions pass; a confound suspends the reading. p2d ran on 2026-08-02 with the instrument frozen exactly as p2c left it, numeric per-stratum predictions committed before any probe was scored, fresh evaluation draws (new salt, filler region, positions), and four arms: the two seen boundaries (Phi-3 at W=2047, Mistral-v0.1 at W=4096), a boundary the fit had never seen (Phi-3 forced to W=1024), and a window-disabled Mistral control. All three windowed arms passed both pre-registered gates — the per-stratum prediction gate (Phi-3: 22 of 24 strata within tolerance against a required 20) and the boundary gate, including at the unseen W=1024. The control arm did not stay flat: with no window enforced, recall declined at extreme absolute distance, triggering the campaign's pre-registered confound branch — if retrieval degrades with distance even without a window, the far-field term in the windowed predictions could be fitting that degradation rather than window arithmetic. The same-day diagnostic p2e reproduced the decline in Mistral-v0.2 — a different RoPE base and extended-context training — so the effect is not an artifact of one checkpoint's positional encoding. The pre-registered interpretation therefore stands at: windowed predictions pass out-of-fit, but the aggregate transfer claim stays suspended until a from-scratch control settles whether far-field decline is a general property of pretrained checkpoints probed zero-shot or something a model trained on the retrieval task also exhibits. That control (p2f) — two from-scratch stablegla arms at an 8,704-token block, one unwindowed and one at W=4096, trained with retrieval supervision at uniformly distributed depths and scored on a frozen battery reaching distance 7,792 — completed as this draft was frozen. Its pre-registered readings were designed to distinguish a flat profile (decline is a checkpoint property; the suspension resolves in the law's favor with the residual attributed to long-context capability) from a declining one (the law's far-field account needs an explicit capability-decay term) — but its precedence rule fired first: the W=4096 positive-control arm did not reproduce a sharp step at this scale, the shape readout came back mixed, and neither branch is taken. p2f is INSTRUMENT_SUSPECT under its own rules, no claim is drawn from it in either direction, and the suspension stands (limitation 14).

p2g: the law used prospectively — a deployed checkpoint is converted, and its recall lands on the sealed prediction. Everything from p2a to p2f asks whether the law describes released checkpoints *as they are*, zero-shot; campaign p2g (`prv-provconv-20260806-p2g`) asks the provisioning question instead: convert a deployed checkpoint to the windowed module, train the conversion under a fixed recipe, and commit — before that training begins — to what its recall will be. The substrate is Qwen3-0.6B-Base at a pinned revision (28 layers, grouped-query attention with 16 query and 8 KV heads, head dimension 128). Four arms trained under an identical two-stage recipe (300M tokens of per-layer attention transfer against the frozen teacher,

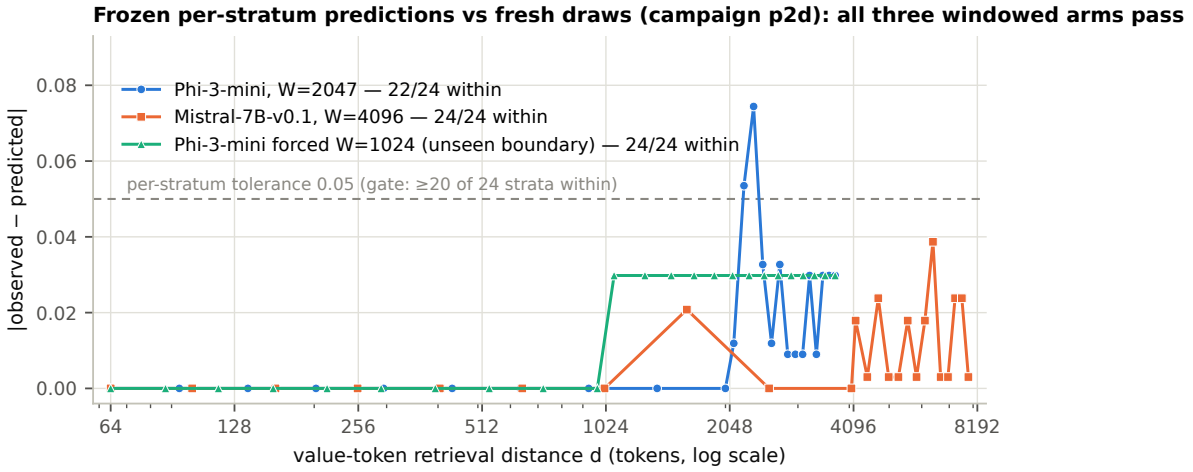


Figure 4: The frozen-prediction pass (campaign p2d). Per-stratum |observed − predicted| for the three windowed arms, with every numeric prediction frozen from p2c's fitted instrument before any probe was scored, on fresh evaluation draws. All three arms pass the pre-registered gate ($\geq$ 20 of 24 strata within 0.05), including at the never-fitted boundary $W = 1024$. The window-disabled control arm (not shown) triggered the campaign's pre-registered confound branch, so the aggregate transfer claim stays suspended (§4.3).

then 200M tokens of LoRA recovery; sequence length 2048; one training seed, 2105; corpus FineWeb-Edu with needle and key–value retrieval formats mixed at 0.10 each — the natural-text supervision regime of §4.2, which we flag now because it bounds what the result can claim): A1, every layer windowed at a declared W=1024, the certification cell; A4, every layer at W=512, co-registered; A2 and A3, folklore layouts at the same W=1024 keeping every third or sixth layer full-attention, as comparators. The registration, gates, per-stratum predictions and battery hashes were committed to the public ledger at 2026-08-06T04:46:33Z, before conversion training started (registered pre-seal smoke tests at the new block length had run real transfer steps; the sealed predictions are pure coverage arithmetic, f = k/24, and cannot be moved by any training outcome); the frozen primary gate reads |observed - predicted| $\leq$ 0.03 aggregate and $\leq$ 0.05 on at least 20 of 24 strata, over a sealed stratified battery (salt 779038744) pooling 9,600 scored retrievals per arm across four certification tasks. Both certified cells pass. **A1: predicted 0.16666674, observed 0.16354167 — an aggregate error of 0.0031, nearly ten times inside tolerance, with 24 of 24 strata within bound. A4: predicted 0.12500007, observed 0.11885417 — error 0.0061, 23 of 24 strata.** The scorer was authored and hashed before either certified arm's scores existed — only the comparators A2 and A3 were complete at authoring, and the certified arms' score files landed after — so it could not have been shaped by the results it adjudicates. The structure is §4.2's step, reproduced through a conversion: in-window strata read 0.920–0.990 and out-of-window strata read exactly 0.000 at every one of the 20 (A1) and 21 (A4) out-of-window strata, n=400 per stratum. Decomposed as the registration requires, r_in = 0.981 (A1, against its registered premise r_in $\geq$ 0.82) and 0.951 (A4), and r_out = 0.000 in both. A4's single out-of-tolerance stratum is its largest in-window distance (d=280, observed 0.920 against 1.0) — degradation nearest the window edge, inside the gate but reported.

The secondary gate — recall per byte of KV state — passes, for a conversion that fails its quality kill-gate (next paragraph); both concessions stated. At evaluation length 8,192 in bf16, a full-attention layer of this model holds 32.00 MiB of KV state; the arms hold 112.00 MiB (A1, 12.5% of the 896 MiB unconverted base), 392.00 (A2), 252.00 (A3) and 56.00 (A4). The break-even recalls were frozen pre-data: A1 beats the 3:1 folklore layout on recall-per-byte iff A2 observes below 0.5833, and the 6:1 layout iff A3 observes below 0.3750. Observed: A2 0.3269, A3 0.1508 — **A1 delivers 1.75× and 2.44× the recall per MiB of its comparators.** Two concessions accompany it. The first we state rather than absorb: on *raw* recall A2 beats A1 (0.327 against 0.164) — its full-attention layers are not window-bound, and A1's claim is that its number was *predicted*, not that it wins raw. The second was registered before data so that it could not be reframed after: A1 was arithmetically guaranteed to lose recall-per-byte to A4 if both certified (f/W falls along this ladder), which is observed (0.00212 against 0.00146 recall/MiB) and is a property of the instrument, not a finding.

The conversion fails its own quality kill-gate, and we put that beside the wins on every reading. The registered KILL gate bounds held-out LM loss against the *unconverted base* — $\Delta \leq 0.10$ nats at sequence length 2048, adjudicated on A1 only. Measured on 262,144 held-out tokens: base 2.5923, converted A1 2.9356 — **+0.343 nats, 3.4 times the budget: FAIL** (+0.356 at length 8,192, reporting-only). The comparison is biased in A1's favour — A1 alone was fine-tuned on the head of the very corpus scored, the base saw none of it — so the deficit is if anything understated. The failure attributes to the conversion — recipe and/or the windowed architecture itself, which this design cannot separate — and not to the law, whose claim is recall only and which predicted recall correctly *through* it. The registered descriptive reporting for the non-certified arms bounds the split: against the same base, the folklore comparators land at +0.318 nats (A2, 10 full-attention layers) and +0.316 (A3, 5) beside A1's +0.343, with the fully-windowed narrower A4 at +0.377 — layouts keeping a third or a sixth of their layers at full attention suffer within 0.03 nats of a fully-windowed conversion, so most of the penalty tracks the shared 500M-token conversion-training recipe rather than the windowing; one seed, descriptive only. The consequence is stated without decoration: certification is a property of a prediction, not of a product, and this converted model is not shippable. The recall-per-byte result above must never be quoted without this paragraph. One instrument note belongs in the permanent record: the per-arm result artifacts carry a flag, `holdout_non_regression: true`, which compares the model after recovery against the model after surgery but *before* recovery — a comparison that training all but guarantees, true by construction, carrying no quality information whatever. The gate above is the only LM-quality statement in this campaign; the flag is an all-clear that cannot return not-clear, of exactly the class §3.6's protocol exists to catch, and it was caught by audit, not by design.

Controls and scoring-path discipline. The random-placement control (A5) was skipped under a pre-registered degeneracy rule — with all 28 layers converted in A1 there is no non-trivial placement to randomize — so KILL(i) is NOT EVALUABLE and is not reported as passed. KILL(iii) passed in its registered form: the converted module initializes as a no-op and reproduces the base model's logits bit-identically on real weights, verified per-arm (maximum absolute difference 0.0 on all four). KILL(iv): every arm ran to its full registered token budget; no efficacy stopping. One training-side amendment postdates the seal and is disclosed here: the first wave-A launch was OOM-killed on all four arms (06:23Z) — the fp32 teacher-plus-student graph exceeded GPU memory — and the operator's repair added gradient micro-batching (micro_batch=1) before the same-morning relaunch. The amendment preserves the registered

data stream exactly: the same seed-and-step-indexed batches, token-weighted mean losses mathematically identical to the full-batch objective, one optimizer step per step, equivalence verified to 1×10^{-9} relative on CPU for both trainers; only peak activation memory changes. The killed attempts are retained in the campaign record (available on request, §Reproducibility; `failed_stage1_*` directories; A1's superseded stage-2 OOM attempt is likewise recorded inside its finalized result under `stage2_superseded_oom`). Mid-campaign, a KV-cache-enabled scoring path was adopted for throughput under a gated amendment procedure whose rule was that an amendment may only *add* ways to fail: 56 held-out probes per arm compared the cached path against the sealed reference — scored answers 56/56 identical, continuation bytes 56/56 identical, production-scorer calls 56/56 identical, maximum logit deviation $\leq 8.5\times10^{-4}$, zero holds, on all four arms.

What p2g does and does not establish. It does not touch the transfer suspension above: the converted model is trained with retrieval supervision in its corpus, which is §4.2's regime — the zero-shot question of p2b–p2f stands exactly where limitation 14 leaves it. It certifies the ceiling at a *declared* window and law-placement against folklore-placement at matched W and budget; it does not certify the law *choosing* W, because the pilot had no deployment distribution to invert against, and any such claim here would be unfalsifiable decoration. The registered sweep-cost accounting is: against the six-window grid the battery guards, a full conversion sweep costs 5× one conversion and a binary search 2×, against 1× for the law path — and the pilot itself ran four conversions of which three are controls a practitioner would not pay. The result is one substrate, one scale, one training seed; the registered follow-on runs only from this pass and has not run: a second seed on a certification pass, and a Llama-3.2-1B substrate, second by signed order. Gaps in the released record are enumerated in limitation 15.

4.4 The compute condition: recall acquisition is bimodal

Across all campaigns (200+ budget-halted runs), per-run hidden-eval recall clusters at ≈1.0 (with final loss 10^{-5}–10^{-8}) or at a non-retrieval level between the uniform-guess rate 0.0625 and the panel's Bayes no-retrieval floor (§4.1.1), never between (loss at the no-recall plateau). On the hard panel family four candidate-architecture runs landed between 0.15 and 0.85 (pooling every arm, including GLA controls at the escalated budgets of campaigns r/t/u, raises that count to 13), and their training logs show the transition in progress at budget exhaustion. (Windowed reads introduce a third, *predictable* attractor — the window-limited ceiling of §4.1 — which is a saturated transition, not an intermediate outcome.) Transition probability varies smoothly with panel difficulty and budget while individual outcomes remain binary — e.g., StableGLA on pairs80 at 3×10^{15} FLOPs transitions with $\hat{p} \approx 0.6$, and the same architecture on the harder p80x384 at the same budget with $\hat{p} \approx 0$. Mean-recall statistics in this regime are mixtures and are not used for claims.

Where bimodality breaks, stated explicitly. The clean bimodality above is a property of the hard panels, and campaigns aw/ax show it does not hold universally. On the easy panel the transformer produced twelve runs inside the 0.15–0.85 band — aw at 0.178–0.406 (8 runs) and ax's non-crossers at 0.410–0.427 (4 runs) — and, unlike the four hard-panel cases, these are **not** transitions caught mid-flight by budget exhaustion: the ax non-crossers hold a training loss of 0.513–0.517 flat from step 40,528 to step 101,320, roughly 60% of training, while their seed-matched siblings at the same budget collapse to loss 0 over the same interval. That is a stable intermediate attractor, not censoring. §4.11's distance-resolved measurement independently confirms it is genuine partial retrieval on this panel (0.26 at distance 0–4, 0.32 at 8–16, 0.56 at

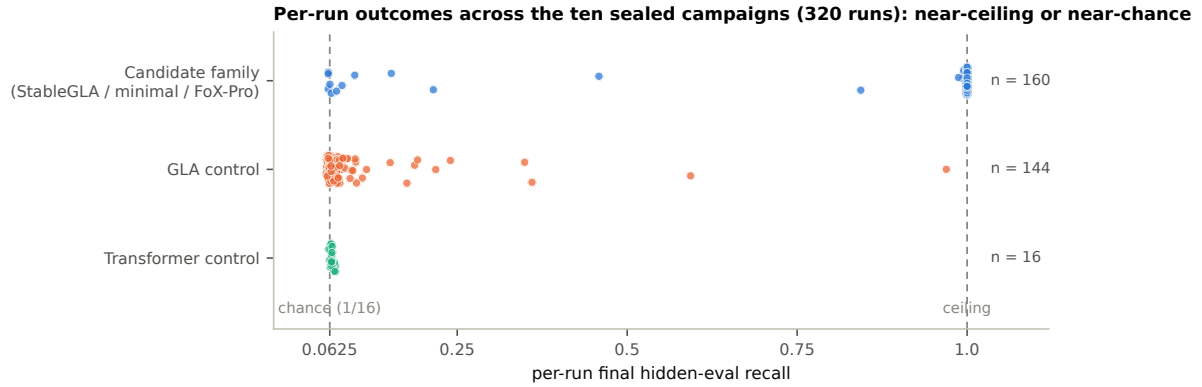


Figure 5: Recall acquisition is bimodal. Per-run final hidden-eval recall for all 320 runs of the ten sealed claim campaigns (m, q, r, t, u, ag, aj, ak, am, aq), by architecture family: outcomes sit at the ceiling or at the 1/16 chance floor. The few runs between are mid-transition candidates caught at budget exhaustion or GLA controls climbing at escalated budget (§4.4).

16–32), where the corresponding `pairs80` profile is flat and therefore reflects no retrieval reach at all. We therefore restrict the claim: on the hard panels difficulty and budget modulate the *probability* of transition and not the attained level, and outcomes are bimodal; on the easy panel a stable partial-retrieval regime exists at intermediate budget and the attained level is itself budget-dependent. Any survival analysis of these trajectories must treat the two panel families separately.

We use "phase transition" descriptively: an abrupt within-run change from chance-level to near-perfect hidden-evaluation recall, together with a bimodal across-seed outcome distribution at fixed finite budget. We do not claim a thermodynamic-limit phase transition, and we note that finite-budget censoring necessarily sharpens the observed bimodality; a censored time-to-transition (survival) analysis over the stored training trajectories is in preparation.

4.5 Sealed confirmation series

Nine sealed campaigns, each with fresh seeds and salt and pre-registered gates, all passed. A tenth sealed claim campaign, aq (FoX-Pro versus transformer, 16/16 against 0/16; §4.10, limitation 3), passed under the same protocol and is pooled in Figure 5; it is excluded from this table because its control arm is the transformer rather than GLA, and it is counted in the claim family of §3.5. Campaign s, run in the same autonomous sequence as r, t and u, was a non-claim budget-escalation probe rather than a sealed claim campaign — its manifest is marked claim-eligible: false — asking whether GLA escapes the shared loss plateau at twice the demonstrated budget: it does not (0/12 GLA transitions at 1.2×10^{16}, against 12/12 for the candidate), and the sealed claim campaigns t and u followed at the budget s had vetted (t's manifest records s as its parent). Campaign s accordingly does not appear in this table; its artifacts are in the released ledger. The first five test StableGLA; ag seals the minimal architecture (§4.6); aj/ak/am — the windowed triple seal — seal the windowed (W=256) minimal architecture across all three task regimes:

Campaign	Candidate arm	Panel	Budget (FLOPs)	Candidate	GLA control	Discordant pairs	**McNemar exact p (one-sided, primary)**	Fisher p (one-sided, unpaired, secondary)
m	StableGLA	pairs80	3×10^{15}	10/16	0/16	10	$\mathbf{9.8\times10^{-4}}$	1.2×10^{-4}
q	StableGLA	p80x384	6×10^{15}	13/16	0/16	13	$\mathbf{1.2\times10^{-4}}$	1.6×10^{-6}
r	StableGLA	seq384	6×10^{15}	15/16	0/16	15	$\mathbf{3.1\times10^{-5}}$	2.8×10^{-8}
t	StableGLA	p80x384	1.2×10^{16}	16/16	0/16	16	$\mathbf{1.5\times10^{-5}}$	1.7×10^{-9}
u	StableGLA	seq384	1.2×10^{16}	16/16	1/16	15	$\mathbf{3.1\times10^{-5}}$	2.8×10^{-8}
ag	minimal arch	pairs80	3×10^{15}	16/16	0/16	16	$\mathbf{1.5\times10^{-5}}$	1.7×10^{-9}
aj	minimal, W=256	pairs80	3×10^{15}	15/16	0/16	15	$\mathbf{3.1\times10^{-5}}$	2.8×10^{-8}
ak	minimal, W=256	p80x384	6×10^{15}	16/16	0/16	16	$\mathbf{1.5\times10^{-5}}$	1.7×10^{-9}
am	minimal, W=256	seq384	6×10^{15}	16/16	0/16	16	$\mathbf{1.5\times10^{-5}}$	1.7×10^{-9}

McNemar p-values are computed from per-seed outcomes under the pre-registered crossing criterion (recall > 0.8); the discordant-pair column makes the effective sample size explicit. Fisher values are the unpaired comparison retained for continuity with earlier drafts. Two further clarifications from the 2026-08-02 methodology review. First, McNemar was not the pre-registered primary: the campaign manifests pre-registered the one-sided Fisher exact bound at $\alpha = 0.005$ as the primary gate, and the promotion of the exact McNemar test to primary is an analysis-level correction made in v0.4 after the pairing argument of §3.4 — the sealed gates themselves were never amended, and every campaign in the table passes under both statistics at its pre-registered α. Second, sidedness: every p-value in the table is one-sided (the McNemar column is an exact one-sided binomial sign test on discordant pairs; the Fisher column is the one-sided exact test), matching the pre-registered directional hypothesis. The released stats-supplement.json reports both sidednesses explicitly and labeled (mcnemar_exact_one_sided_p, mcnemar_exact_two_sided_p); its one-sided values match this table exactly.

The single GLA transition (campaign u, 4× the candidate's sufficient budget) is informative: GLA's threshold is finite but far higher — consistent with a compute-threshold account rather than a categorical incapacity.

4.6 Mechanism I: the component matrix and the minimal architecture

Single-line ablations of StableGLA (6 seeds each, pairs80@3×10^{15}; parameters and FLOPs preserved by zeroing rather than removing modules) began the matrix: collapsing the log-spaced half-life prior to a constant leaves the transition intact (5/6); removing softmax normalization of the read leaves it partially intact (3/6); zeroing the input-dependent decay modulation yielded 0/6; a second subtractive round found removing the short depthwise convolution also 0/6, removing the softplus content kernel *increased* the rate (6/6), and removing the write gate or

QK-normalization reduced it (3/6 and 2/6 with partials). We describe components as "necessary" strictly in this configuration-bounded sense (n=6 per arm).

Additive reconstruction then identified the *minimal sufficient architecture*: a bare core (convolution + input-dependent decay bias + softmax read) does not transition (0/6), nor does the core plus any single supporter (softplus 0/6, write gate 0/6, QK-norm 1/6, output-norm 2/6 with partials) — but core + QK-norm + output-norm transitions 6/6 at both 3×10^{15} and 6×10^{15}, and was then *sealed* against fresh paired GLA controls: 16/16 vs 0/16, Fisher p = 1.7×10^{-9} (campaign ag; table in §4.5). An earlier, different construction — the write-gated positive-kernel "minimal mixer" of the first hardening round, which transitioned 3/6 — is a distinct architecture and should not be conflated with this sealed minimal architecture; we retain it in the record (Appendix C, campaign p) as a superseded intermediate.

Two further dissociation controls complete the picture: a transformer given the same short convolution but no decay (attnconv) does not transition (0/6, two partial runs; an adverse single-layer conv-rescue result in the literature is engaged in §4.10), and GLA — which has input-dependent decay and a convolution but no softmax read — is 1/96 throughout (96 pools the 80 paired GLA controls of the five StableGLA sealed campaigns, §4.5, with the 16 GLA runs of the learning-rate sweep, §4.8; across the paired GLA controls of all nine sealed campaigns the count is 1/144, as in the abstract). The pattern across all arms is a **three-way dissociation**:

1. **Softmax read without decay** (transformer, attnconv): 0/182 and 0/6 on the hard panels — does not cross at budgets up to 44× (at least fortyfold) the sufficient budget. This is a threshold statement, not a categorical one: on the easy panel at 3×10^{16} FLOPs (10× the candidate's sufficient budget there) the same transformer crosses in 2/6 seeds (§4.10), so the dissociation below is bounded by compute and panel and should be read as such.
2. **Input-dependent decay without a softmax read** (GLA): 1/96 — does not cross except once at 4× budget.
3. **Softmax read + input-dependent decay + local KV preprocessing**: crosses reliably at 3×10^{15} (StableGLA, the minimal architecture, and — §4.7 — every gate-form variant and the published FoX-Pro assembly).

Within regime 3, the *form* of the decay gate is immaterial and the *presence* of preprocessing is decisive; the normalizations are transition-rate boosters (§4.7).

4.7 Mechanism II: why the Forgetting Transformer's Pro block works

Version 0.3 of this paper claimed that the published FoX formulation did not transition and that no single- or two-component augmentation rescued it. Campaigns an and ao (2026-07-30) show that claim was wrong in an instructive way, and we correct it here in full.

The *base* FoX formulation (cumulative log-sigmoid forget bias on standard scaled-dot-product logits; no preprocessing, no normalizations) indeed does not transition: 0/22 across the original baseline and a gate-bias-init × positional-encoding sensitivity sweep, and 0/48 further runs adding one or two StableGLA components (convolution, softplus kernel, prior, write gate, pairwise combinations) never reached the full shell. But:

Arm (6 seeds each, pairs80@3×10^{15})	Transitions	Reading
FoX-Pro exact (published Pro: QK-norm + output-norm + sigmoid output gate + data-dependent KV-shift + unbounded gate)	**6/6**	the published assembly crosses
FoX-Pro minus KV-shift	0/6	preprocessing is necessary
Shell (conv + norms) with FoX's unbounded gate	6/6	gate boundedness is immaterial
Shell with one-sided σ(u) gate (Griffin/HGRN-style)	6/6	gate two-sidedness is immaterial
Shell minus convolution (= FoX + norms)	0/6	preprocessing is necessary
Shell minus QK-norm	3/6	norms are rate boosters
Shell minus output-norm	2/6	norms are rate boosters
Minimal-architecture reference (concurrent fresh seeds)	6/6	control

The dissociation of §4.6 therefore extends cleanly into the FoX family: what separates base FoX (0/22) from FoX-Pro (6/6) in this compute regime is *local preprocessing of the KV path* — Pro's data-dependent KV-shift plays exactly the role of our short convolution (either suffices; removing either yields 0/6) — with QK- and output-normalization lifting the transition rate from partial (2–3/6 with one norm, 0/6 with neither) to reliable (6/6). Our convolution additionally preprocesses the query path, which the KV-shift does not; the equivalence of the two arms locates the shared necessary component on the KV path, but a Q-only/KV-only factorization of the convolution has not been run. The specific decay parameterization — FoX's unbounded log-sigmoid, a one-sided gate, or our bounded two-sided modulation — makes no measured difference. The FoX paper reports that Pro outperforms the base instantiation and ablates the Pro components individually (its Table 3) but measures only perplexity and does not attribute the gap mechanistically. To our knowledge this is the first identification of which Pro components are load-bearing for recall acquisition specifically, on an endpoint where the KV-shift moves from a 0.18-perplexity-point contributor (6.62 → 6.80 in FoX's Table 3, ≈0.027 nats, under three percent relative) to a precondition. These mechanism campaigns are non-claim (n=6 per arm) and configuration-bounded; the sealed FoX-Pro-vs-transformer claim campaign (aq) has since completed and passed (16/16 vs 0/16; §4.10 and §5, limitation 3). At n=6 per arm, a 0/6-versus-2/6 contrast is not by itself significant; the preprocessing claim rests on its replication across every arm of the matrix (0/6 in all configurations lacking it, 0/22 base FoX, 0/48 augmentation runs), while the norms-as-rate-boosters reading is descriptive.

4.8 Robustness: learning-rate sweeps

All preceding failures used base_lr = 3×10^{-3} (AdamW, warmup–cosine). Following the optimization-artifact critique [2508.19029], we swept base_lr ∈ {3×10^{-4}, 10^{-3}, 6×10^{-3}, 10^{-2}} with 4 fresh seeds each, identical seeds across arms: transformer 0/16 (mean recall 0.126, 0.115, 0.064, 0.061 respectively — low rates lift recall above the uniform-guess rate to approximately the Bayes-optimal

no-retrieval accuracy for this panel (0.1260, §4.1.1) — that is, to the ceiling of not retrieving, not part-way toward transition), GLA 0/16. The inversion is not a learning-rate artifact at the sweep's resolution. Two caveats we state rather than bury. First, the transformer's argmax sits at the *lowest* rate tested, so the axis is not formally bracketed downward; the increments decay ($0.064 \to 0.115 \to 0.126$), and we can now account for that value rather than merely note it: 0.1260 is the Bayes-optimal accuracy attainable on `pairs80` without performing any retrieval at all (§4.1.1). The decaying increments are not an approach to the 0.8 transition threshold interrupted at an arbitrary value; they are convergence on the ceiling of not-retrieving. Campaign as extends the sweep to 10^{-4} and to budgets of 4.8×10^{16} to close it. As's own pre-registered meta-gate on this point *fired* (`meta_gate_sweep_unbracketed: triggered = true`, argmax still at the lowest rate tested, consequence recorded as "a further downward LR extension is REQUIRED before publication"); campaign au ran that extension and closed it, finding the rate turns over rather than continuing to improve. We also state the shape of the coverage rather than let it be inferred: the LR and budget axes are two marginal lines, not a factorial grid. The 4.8×10^{16} cells were run only at base_lr $\in \{10^{-4}, 3\times10^{-4}\}$, and the LR optimum is panel-dependent — on `pairs80` the transformer's best rates are 10^{-4}–5×10^{-4} (recall $\approx$0.13) while 3×10^{-3} sits near chance, yet 3×10^{-3} is the rate at which it crosses on the easy panel. A sweep over one axis at fixed values of the other therefore does not cover the space, and the corner (3×10^{-3}, $\geq4.8\times10^{16}$) on `pairs80` remains untested. Second, the flatness itself is the predicted behaviour: [2508.19029] finds standard transformers robust to learning rate and modern recurrent models acutely sensitive, so a transformer that does not respond to a 33× change in rate is the expected result, not an anomaly.

4.9 The window boundary on synthetic panels: an instructive failure

We attempted to locate the window boundary empirically on the synthetic panels — is the last attendable distance W or W-1, and does stacked-layer relay extend it? — and the attempt failed in a way worth reporting in full, because the design that finally settled the question (§4.2's Battery B on natural text, and the structural mask checks of §4.3 on deployed checkpoints) is the one this failure forced. Campaign ar ran on a panel where every retrieval distance is exactly 160, and **its result does not support a conclusion about reach in either direction.** Transition counts by window were: W=156 (direct reach 155) 1/6; W=158 (157) 6/6; W=160 (159) 5/6; W=161 (160) 2/6; W=162 (161) 3/6; W=164 (163) 0/6. The last cell is decisive against any reach-based reading: W=164 covers the target outright with three tokens to spare and yet failed in all six seeds at a maximum recall of 0.126 — precisely this panel's no-retrieval floor (§4.1.1), i.e. the runs recover nothing whatever — on runs that are otherwise clean (identical token counts, state-tracking solved at 1.0, losses on the plateau that marks non-acquisition). More available context produced total failure while less produced perfect success. Window width in this design governs *optimization difficulty* — a wider window admits more candidate positions to suppress at a fixed compute budget — and difficulty is confounded with reach. This is not a novel effect and we should have anticipated it: SWAX [arXiv 2509.24552] reports that "larger sliding windows hurts the long-context performance" and attributes it to weakened optimization pressure when a wide local window already covers the useful dependencies. The receptive-field bound for stacked sliding-window attention (reach $\leq$ window $\times$ depth) is likewise standard, so the depth-relay hypothesis ar registered was a test of a known bound rather than a new proposal, and the campaign was not designed to separate it from the difficulty effect. Transition count measures reach only if difficulty is held constant, and it is not. We therefore report the *synthetic*

boundary question as **open**, and on these data neither the strict nor the inclusive convention nor a depth-relay account is supported. Campaign at inverts the design — hold the window fixed and sweep the retrieval distance across the boundary, vary the convolution kernel and block count independently, add a jittered-distance control, n≥16 with a graded statistic — since only that separates the two; it is registered, and its outcome is reported below. On natural text the inverted design *has* been run (§4.2): retrieval is intact at distance exactly W and collapses at W+1, settling the convention as inclusive for our implementation; on released checkpoints the structural boundary is exclusive at W-1 (§4.3). The synthetic question remains **open**, and campaign at has since reported without settling it: it completed 44/44 and returned INSTRUMENT_SUSPECT. Two independent reasons are now identified. First, its panel carries a no-retrieval floor of 0.4235 (§4.1.1), so sub-threshold recall on it measures guessing rather than partial reach. Second, the stacked-window receptive field L·(W-1)+1 was 53–157 across its depth arms while the battery's largest observable distance was 32, so no collapse could have been observed even from a clean panel. A repaired-instrument attempt (`instrument-repair-20260811-au` — a second campaign bearing the suffix au, distinct from and unrelated to the learning-rate bracket campaign of §4.8; the suffix collision sits in the released ledger and is disclosed here, alongside the campaign-i collision of Appendix C) failed its own learnability gate. Both of those obstacles have since been removed by construction, and the question does not move. A purpose-built panel, `permval16`, draws its values as a permutation with P = NVAL = 16 and scores a single probe at query ordinal 0, which makes the un-asked pool a constant multiset carrying no information and fixes the no-retrieval floor at exactly 1/16 analytically rather than at easy48's 0.4235; and at W=14 with L=6 the stacked receptive field L·(W-1)+1 = 79 covers the entire observable band d ∈ [32, 76], so no distance in the battery lies beyond reach. Four seeds at L=4 and four at L=6 nonetheless returned mean aggregate recall of 0.0649 and 0.0645 against the 0.0625 uniform rate, each halted by a pre-registered early abort at 45,000 of 180,000 optimizer steps with the co-trained state-tracking task solved at ≈1.0 — so the harness trains and the failure is the task's, not the pipeline's. These are claim-ineligible learnability probes on scratch salt and no claim in this paper rests on them. We therefore report the synthetic reach question as open not merely for the two reasons above but after both were removed, and propose no further attempt here: on this evidence the measurement requires more scale than this programme commands. The provisioning guidance is unchanged and unaffected — size W by the strict convention and treat any extra reach as margin, which is conservative under every hypothesis including the null. One structural fact, verified from the released harness, belongs in this record: the candidate's short causal depthwise convolution (k=4, applied before the QKV projection) folds each position's three predecessors into its keys and values, so the oldest attendable position under the strict mask (direct reach W-1) carries raw-token content from up to W+2 back — for W=158 that is exactly 160, the panel's fixed prediction-position-to-value distance, which suffices to explain the low-W successes without any multi-layer relay (only the single W=156 transition, at single-layer reach $158 < 160$, requires one), while leaving the high-W failures to the difficulty effect above. This does not contradict §4.2's collapse at W+1: the boundary batteries there measure the trained model end-to-end with the same convolution included and under a different distance anchor (distance is measured from the answer position to the planted needle, one more than from the prediction position that must retrieve it, so §4.2's "intact at exactly W" is the same structural reach of W-1 attendable positions), and what §4.2 shows is that on natural text recall steps at the mask edge — the convolutional reach beyond it, which this fixed-distance panel forces training to exploit, goes unexploited when the training distribution never requires it.

4.10 Where attention's threshold lies, and where it does not

On pairs80 at 3×10^{15}: transformer 0/8, hybrid 0/8. Budget escalation on the same panel: transformer 0/3 at 6×10^{15} and 0/3 at 1.2×10^{16} — four times the budget at which the candidate transitions with $\hat{p}\approx0.6$, and twice the budget at which it transitions deterministically on a *harder* panel. With the learning-rate sweep of §4.8 the transformer total was 0/30; the sealed claim campaign aq (§4.5) adds 58 further transformer runs — 16 paired at 3×10^{15}, a learning-rate $\times$ budget grid at 6×10^{15} and 1.2×10^{16}, and a token-matched arm — all of which returned 0 transitions, bringing the transformer total on this panel family to 0/88 at v0.4.1. Campaigns as (56 runs), au (24), the interior probe av (8) and the step-matched control az (6) have since added 94 further transformer runs, all non-transitioning, for a current total on this panel family of **0/182 — 182 runs on 170 distinct seeds across 27 distinct panel $\times$ budget $\times$ learning-rate cells, spanning 2.8×10^{15}–1.32×10^{17} FLOPs (the learning-rate sweep reaches 4.8×10^{16}; the 1.32×10^{17} cell is az's single step-matched rate) and base learning rates 10^{-6}–10^{-2}, at a maximum observed recall of 0.1365** — against a Bayes-optimal no-retrieval accuracy of 0.1260 on this panel family (§4.1.1), that is, the best of 182 transformer runs exceeds what is achievable without any retrieval by 0.011, and no run reaches one-sixth of the 0.8 transition criterion. The pooled distribution is bimodal about these two levels (67 runs below 0.07, 67 in [0.126, 0.137], none above 0.14). We therefore no longer describe the non-crossing runs as "at chance": a substantial minority are at the ceiling of not-retrieving, which is a stronger and more specific statement about what attention does instead. The learning-rate span carries one qualification, previously disclosed only in the version history and stated here where the claim is made: the rate is attributable from released run artifacts for 138 of the 182 runs; the remaining 44 (aq 22, n 8, av 8, o 3, p 3) record no learning rate in any artifact — they are pooled in the run count but excluded from the 10^{-6}–10^{-2} rate-coverage claim, and the recount script (released in the ledger repository, §Reproducibility) reports them explicitly rather than silently absorbing them. All counts in this paragraph are recomputed from the released run artifacts by the script cited in Reproducibility; earlier drafts reported 88, 168 and 172 at various points and those figures are superseded. Runs and seeds differ because campaign w deliberately reuses four seeds across its four learning rates (a paired sweep), so the pooled count is reported as runs, distinct seeds and cells rather than as 182 independent draws.

Two limits of this coverage were stated in earlier versions and **campaign az has now closed both.** First, the sweep was two marginal lines rather than a factorial grid: the 4.8×10^{16} cells were run only at base_lr $\in \{10^{-4}, 3\times10^{-4}\}$, and the harness-default 3×10^{-3} — the rate at which the transformer is observed to cross on the easy panel — had never been run above 1.2×10^{16} on `pairs80`, leaving an untested corner at precisely (3×10^{-3}, $\geq4.8\times10^{16}$). az is that corner: `pairs80`, base_lr 3×10^{-3}, 1.32×10^{17} FLOPs. Second, and more consequentially, matched measured FLOPs buys the quadratic read fewer optimizer steps on a 320-token panel than on a 48-token one: `easy48` at 3×10^{16} affords 101,322 A-phase steps, whereas `pairs80` at 4.8×10^{16} afforded only 21,797, below the 40,528–60,792 steps in which the easy-panel transition is observed. az was sized to **59,943 steps** specifically to span that window, and it is now the largest budget anywhere in this programme. Its pre-registered reading was fixed before any data: 0/6 would be `NULL_SURVIVES_STEP_MATCH`, $\geq$1/6 would be `NULL_IS_STEP_BOUNDED` and would have qualified the transformer claim. The result was **0 of 6** (recalls 0.1201, 0.1143, 0.1139, 0.1130, 0.0737, 0.0736; final losses 0.938–0.987, all on the non-acquisition plateau). We therefore no longer report the hard-panel result as inseparable from an optimization-time bound: at the one learning rate known to work on this architecture, and at a step count that brackets the observed transition,

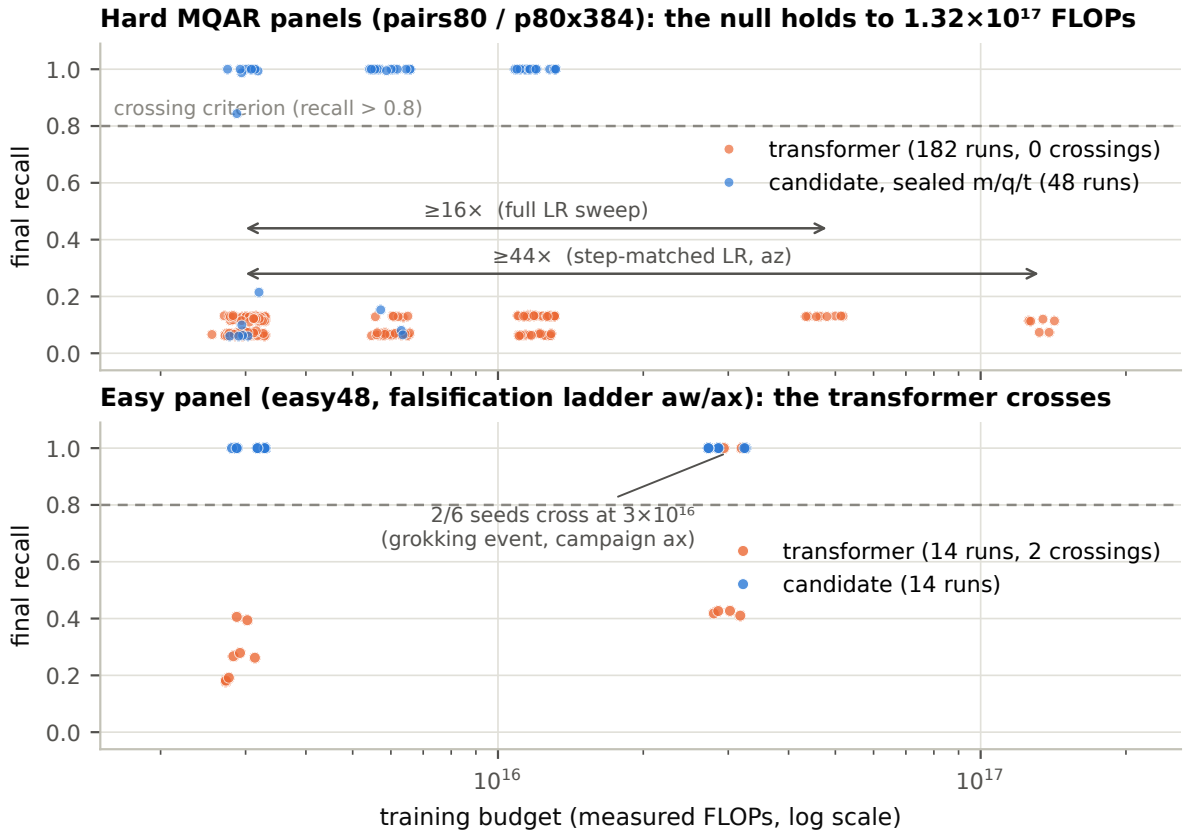


Figure 6: Where attention's threshold lies, and where it does not. **Top:** on the hard MQAR panels the transformer crosses in none of 182 runs (recomputed from run artifacts by the released recount script) at budgets to 1.32×10^{17} FLOPs, while the sealed candidate transitions from 3×10^{15} — a tested separation of at least sixteenfold across the full learning-rate sweep and at least fortyfold at the step-matched rate (campaign az). **Bottom:** on the deliberately easy panel the same transformer crosses — 2 of 6 seeds at 3×10^{16} FLOPs, a textbook grokking event (campaign ax) — so the null is a threshold claim, not an incapacity claim.

the null holds.

The falsification test we ran against ourselves. Because a null of this size invites the objection that the harness simply cannot produce a working transformer, we pre-registered an escalation ladder (campaign aw, with the budget rungs and stopping rule fixed in `campaign.json` before any data): Stage 1 at $3{\times}10^{15}$ on 8 paired seeds; if the transformer crossed in $\geq$1 seed the harness is validated and the campaign stops; if 0/8, Stage 2 at $3{\times}10^{16}$ on 6 fresh seeds; if 0/6, Stage 3 at 10^{17}. The panel was deliberately eased to approximate Zoology's reported regime (48 tokens, 8 pairs, 64 keys). Stage 1 returned transformer 0/8 (recall 0.178–0.406) against candidate 8/8, McNemar exact one-sided $p = 3.9{\times}10^{-3}$, mechanically triggering Stage 2. **Stage 2 returned transformer 2/6** — seeds 2211 and 2212 reached hidden-eval recall 1.000 on both independent eval draws at final loss $1.6{\times}10^{-9}$ and $6.8{\times}10^{-8}$ — against candidate 6/6 (all six seed-matched candidates reached recall 1.000). Per the ladder's own pre-registered rule the campaign stopped there; no Stage 3 was run, and we did not escalate after seeing a partial crossing. The crossing

and non-crossing runs are byte-identical in source hashes, eval salt, panel, budget, token count and parameter count, and seeds 2210 (no cross) and 2211 (cross) ran sequentially in the same process on the same GPU, so the difference is the seed and nothing else. The loss trajectories show a textbook grokking event: flat at ~0.520 through step 40,528, then $\leq$0.003 by step 60,792. We report two consequences plainly. Attention acquires MQAR under this harness, so the null is not an instrument artefact — this was the single largest open weakness in the previous draft. And the Stage-2 paired statistic is not significant: 6/6 against 2/6 gives four discordant pairs and a McNemar exact one-sided p = 0.0625, above 0.05 (campaign `decision.json`, completed 2026-07-31T20:56:15Z). At 3×10^{16} on the easy panel the transformer–candidate difference is no longer statistically significant at n=6, and we state that ourselves rather than leave it to be derived. A further limitation of the ladder's design: the two budget rungs use different eval salts and disjoint seed blocks, correctly per the sealed protocol but meaning the 3×10^{15}-versus-3×10^{16} contrast is unpaired in both seed and eval draw, and it should not be read as a paired budget comparison.

An adverse conv-rescue result in the literature, engaged rather than waved away. Okpekpe & Orvieto [2508.19029] — cited below for the induction-head-like loss bump — also report a result in direct tension with our attnconv arm: incorporating "a 1D convolution before the Query, Key and Value matrix projections" enables a *single-layer* attention model "to solve MQAR with just one layer", where single-layer attention without it fails; convolution on the key or value projection alone suffices, while convolution on the query alone does not. Our attnconv arm — the same short convolution ahead of the QKV projections, on an 8-block transformer with no decay — is 0/6 on the hard panels (§4.6). The candidate reconciliations are the protocol differences already named for Zoology: their models are trained for 50 epochs with performance reported at the best learning rate over a per-model grid, on MQAR panels reaching 512 tokens/64 pairs over an ~8k vocabulary, while our arm halts at a fixed measured-FLOP budget on 320-token/80-pair panels at one depth and one rate; a kernel-width or placement difference cannot be excluded either. One element of their result reads as corroboration rather than tension: conv-on-Q-alone failing while conv-on-K-or-V-alone suffices matches §4.7's localization of the necessary preprocessing to the KV path — the factorization our own matrix has not run. But the conjunction — conv+attention sufficient at converged training on easier panels, conv+attention 0/6 under budget-halted training on hard panels — has not been tested within a single design, and nothing here rules out attnconv crossing at some budget between ours and convergence. Settling it requires a dedicated arm (attnconv under the aw/ax-style escalation ladder), which the released harness supports and which we have not run.

Where the failure lives (campaign av). A pre-registered probe asks whether a non-transitioning transformer nonetheless *contains* the answer. Eight seed-matched pairs (transformer and candidate, seeds 2200–2207, p80x384 at 6×10^{15}) were trained and each model's residual stream was read at the query position by a linear classifier fitted on disjoint sequences, targeting the correct value token (chance = 1/16 = 0.0625; the Bayes-optimal no-retrieval accuracy on this panel is 0.1260, and the transformer probe reaches neither). A shuffled-label control was fitted on the same activations for every model as a validity gate; three candidate models fell outside the pre-registered gate of chance $\pm$ 0.02 and were excluded rather than explained, as pre-registered. Results: candidates crossed 8/8 and transformers 0/8 (McNemar exact p = 3.9×10^{-3}), reproducing the positive control. Transformer probe accuracy reached 0.067–0.073 across all eight models and all eight layers, against shuffled-label controls of 0.058–0.070 on the same activations — no recoverable signal. Candidate probe accuracy was 0.977–1.000. Among

the five pairs where both members passed the validity gate, the candidate probe exceeded the transformer probe in 5/5 (sign test $p = 0.031$), with an effect size of roughly 0.99 versus 0.07. The probe therefore demonstrably detects the circuit when it is present, and finds nothing when the transformer fails.

The calibration av could not do, done (campaign ay). Every transformer av probed had *failed* the task, so its instrument had only ever been calibrated across architectures — a candidate that solves the task against a transformer that does not. That leaves an obvious objection: perhaps the probe simply reads attention-shaped representations poorly, and "the circuit is absent" is an artefact of architecture rather than a fact about the model. Campaign ax supplied the missing case, transformers that solve the task, and campaign ay put the identical probe on them. In the two crossing transformers (seeds 2211, 2212, recall 1.000) probe accuracy reaches **0.7126 at layer 4 and 0.8248 at layer 7**, against shuffled-label controls of 0.0758 and 0.0541 on the same activations and a chance rate of 0.0625; both models pass the pre-registered validity gate and none were excluded. The same probe, on the same architecture, at the same budget, reads 0.07 when the model fails and 0.71–0.82 when it succeeds. The instrument is therefore validated within-architecture, and av's reading — that failure is representational rather than a readout problem — is supported rather than assumed. One asymmetry is worth recording instead of smoothing over: the crossing transformers probe at 0.71–0.82 while the seed-matched candidates reach 1.000 at layer 0, so even a transformer that has solved the task carries the answer less linearly accessibly, and deeper, than the candidate does. We do not have an account of that difference and do not offer one.

Flat loss is weak evidence of nothing being learned. Our transformer runs plateau at a training loss of ~0.988 from roughly step 272 to step 1362 and evaluate at or below the panel's Bayes no-retrieval floor (§4.1.1) rather than at the uniform-guess rate. We report that as non-acquisition within budget, but note the limit of the inference: Quiet Feature Learning [arXiv 2505.03997] shows validation loss remaining flat over extended compute ranges while causally necessary intermediate features are already forming, and [2508.19029] reports attention on AR producing an induction-head-like loss bump that yields no accuracy gain. A flat loss curve is therefore consistent both with a circuit that never forms and with one forming quietly beneath the metric. Distinguishing these requires reading the trained model rather than its loss, which we have not done; a probe for induction-head structure on a trained transformer checkpoint is the obvious next measurement and we flag its absence as a limitation.

The tension with Zoology, stated plainly. Zoology reports softmax attention solving MQAR at a constant model dimension of 64 across sequence lengths, and it is the reference result in this literature. Our transformer is six times wider (d=384, 8 layers, ~14M parameters) and does not solve the task in 182 runs on the hard panels. We do not claim attention *cannot* do MQAR, and any reading of this paper as such a claim is a misreading. The two results differ in protocol, not in architecture: Zoology trains to convergence and asks whether the circuit is *learnable*; we halt every arm at a fixed measured-FLOP budget and ask whether the circuit is *acquired by that budget*. Those are different questions, and only the second is ours. **Previous versions of this paper flagged a reproduction of Zoology's converged setting — confirming attention succeeds under our harness — as the cleanest way to close this gap, and recorded it as not yet done. It has since been done, and it succeeded.** Campaigns aw/ax, described above, put the transformer on a Zoology-like easy panel and escalated budget along a ladder fixed before any data: attention crossed in 2 of 6 seeds at 3×10^{16} FLOPs, at recall 1.000. The honest statement of our result is therefore narrower and better specified than

before: on the hard panels the budget at which attention acquires MQAR is above 4.8×10^{16} FLOPs across the learning-rate sweep (above 1.32×10^{17} at the step-matched rate) and remains unmeasured, while the candidate's is at or below 3×10^{15} — a separation of at least sixteenfold, and at least fortyfold at the step-matched rate, with the upper end unknown; on the easy panel both ends are now measured, at a separation of at least tenfold. Attention's threshold is finite, it is architecture-dependently higher, and on one panel we have found it.

4.11 Reading the trained model: what the transformer does instead

Sections 4.10 and 4.8 establish that the transformer does not cross on the hard panel family, across 182 runs, a 10^{-6}–10^{-2} learning-rate range and budgets to 1.32×10^{17} FLOPs (to 4.8×10^{16} across the full learning-rate sweep). They do not establish *why*, and a flat loss curve is weak evidence of nothing being learned (Quiet Feature Learning, arXiv 2505.03997). We therefore trained models under the sealed harness and inspected them directly. The training loop is reproduced from `train.py` rather than modified, since `train_run` does not return the model and the file is under hash seal; the reproduction is validated by requiring final recall to reproduce the sealed campaigns on the same seeds (0.1275 against `as`'s 0.1277; 1.0000 against `aq`'s 1.0000 for the candidate; 0.0663 against `aq`'s 0.0633, the last pair both at the chance floor and differing within GPU non-determinism, which we report rather than claim bit-exact reproduction).

A calibrated mechanistic probe. For each layer and head we measure the attention mass the query position places on the answer token — the induction-head score under this panel layout, since the answer sits immediately after the matching key. The measurement is meaningless without knowing what a *working* circuit looks like, so the candidate supplies the calibration:

model	recall	attention mass on the answer token	× uniform baseline
candidate (FoX-Pro-class shell)	1.0000	**0.90**	187×
transformer, easy panel	0.4235	0.07	1.7×
transformer, `pairs80`	0.1298	**0.01**	2.0×
uniform attention over the prefix	—	0.005	1×

A model that solves the task routes ~90% of one head's attention mass onto the retrieval target, and almost none onto the key itself (0.001–0.004) — consistent with the KV-shift folding the key into the value position's representation. The transformer routes 1–2%, barely distinguishable from attending nowhere in particular.

This metric is withdrawn, by our own mechanism campaign, and no claim in this paper rests on it. Campaign pr1 (2026-07-31) read the trained models directly and recorded the attention-mass metric as *uninformative as constructed*: within the transformer arm it runs **anti-correlated with capability** — the failing hard-panel model carries 4.0× the uniform baseline while the better-performing easy-panel models carry 1.6× and 2.3×. pr1's own record states the consequence plainly: "An earlier reading of these same runs claimed the transformer never forms a retrieval circuit. The attention-mass metric cannot support that claim in either direction; it is withdrawn." Earlier versions of this section nonetheless leaned on the

candidate-versus-transformer contrast in the table above (187× against 1.7×) as if the metric were sound across architectures while conceding it was noise within them. That position is not tenable: a metric that inverts against capability inside one arm cannot be trusted to rank across two. The table is retained as a record of what was measured, not as evidence.

The load-bearing evidence that the failing transformer has not built the retrieval circuit is therefore (i) the interior probe of §4.10, now calibrated within-architecture by campaign ay, and (ii) the distance-resolved accuracy below — both of which survive pr1's retraction untouched. The scope of that claim is load-bearing and we have tightened it: every checkpoint in the table above is a non-crossing run, and campaign ax has since produced transformers that *do* cross on the easy panel at 3×10^{16} FLOPs. The analogous calibration for the *interior* probe of §4.10 has since been run on those crossing models and it passed decisively (campaign ay: 0.71–0.82 against 0.07 in failed models). **The attention-mass metric in the table above has not had that treatment**, and we distinguish the two rather than let the one borrow credibility from the other: ay probed linear decodability, not attention mass, and its models were probed in-process and then discarded, so no checkpoint survives to measure attention mass on. Doing it requires one more training run with the mass probe attached. That measurement would now adjudicate the withdrawn metric rather than draw on it: no genuinely crossing transformer has ever been measured (pr1's better-performing easy-panel models sat at 0.42, well short of the 0.8 criterion), so a crossing model reading ~0.9 would rehabilitate the metric and a crossing model reading 1.6–2.3× would confirm its retraction. Until that is run, the table's transformer rows are non-crossing models only and nothing is claimed from them.

Accuracy resolved by retrieval distance. The behavioural measurement agrees. On `pairs80` the candidate answers correctly at **1.000 in every distance bucket** from 4 to beyond 128. The transformer's accuracy is *flat* — indistinguishable at distance 8 and distance 200:

distance	8–16	16–32	32–64	64–128	128+
candidate	1.000	1.000	1.000	1.000	1.000
transformer	0.125	0.000	0.047	0.116	0.129

Its above-chance score is therefore **not partial retrieval at short range**: the model has no effective retrieval reach at all rather than a small one, and the ~0.13 plateau of §4.8 must not be read as the transformer nearly doing the task. On the easy panel, by contrast, transformer accuracy *is* distance-structured (0.26 at 0–4, 0.32 at 8–16, 0.56 at 16–32), so attention performs genuine retrieval when the task is within reach and the harness is capable of producing the circuit. The failure on `pairs80` is not an artefact of the setup.

The same question asked of a real pretrained model. The probes above are synthetic. We also scored an already-trained natural-language checkpoint from the S0.5 build (the program's internal name for the small real-text training tier) — a compute-optimal 110M-parameter transformer at 2.5B tokens, step 78126, unrestricted attention, scored on held-out batteries. It acquires local copying at 0.875 (every retrieval distance 7 tokens) while in-context key–value retrieval sits at 0.008 (distances $\leq$27) and needle retrieval at **0.000** (distances 30–915); induction scores 0.021 and state tracking 0.125. An independent re-implementation of the scorer reproduced 0/32 on the key–value battery, and decoded examples confirm the tasks are well posed, so this is a capability result rather than a scoring artefact. Note that a naive induction head predicts the token that *followed* the key — "is" in "Fact: yellow is Monday" — not the value, so key–value retrieval requires a strictly harder circuit than induction. This is independent evidence, on real

text at 110M scale, that retrieval does not arrive free with generic language competence. It also halted the first law-transfer attempt: a battery in which the long-distance family is unlearnable and the learnable family has a single fixed distance of 7 tokens has no dynamic range for a window study, since every candidate window covers 7 and every predicted ceiling collapses to 1.0. That diagnosis is what motivated training the retrieval format explicitly in the transfer design (§5, limitation 1).

A residual we could not explain — and have now explained. On the easy panel the transformer reaches only 0.42 at 6×10^{15} FLOPs, well short of the near-perfect MQAR accuracy Zoology reports for attention. We pre-registered and tested the obvious explanation — that our frozen suite's 70/30 MQAR/state-tracking mixture starves the retrieval signal, since state positions dominate the masked-token loss. Trained on **MQAR only**, with state removed entirely, the transformer lands at 0.1298 on `pairs80` (against 0.1275 mixed) and 0.4209 on the easy panel (against 0.4235 mixed). **The mixture is not the cause.** Earlier versions of this paper recorded the residual as unexplained. It is now measured: at ten times the budget on the identical easy panel, the transformer reaches recall 1.000 in 2 of 6 seeds (campaign ax, §4.10), with the loss trajectories flat at ~0.520 through step 40,528 and collapsed below 0.003 by step 60,792. **The 0.42 plateau is budget-limited, not a harness pathology** — the MQAR-only ablation above now reads as further confirmation that the mixture was never the cause, since removing it changed nothing while adding compute changed everything. The remaining open question is not why the transformer plateaus but why it plateaus at a *stable* intermediate value for tens of thousands of steps before either transitioning or not; see §4.4, where this behaviour qualifies the bimodality claim. Readers should treat the absolute transformer numbers as properties of this harness at a stated budget, and note that the candidate is subject to every one of the same conditions and transitions 6/6 at a tenth of the compute.

4.12 No observed trade-off on the frozen suite

From sealed-campaign rows (no additional compute): state-tracking 1.0 for all arms; continual-learning retention higher for the candidate — in campaign q, mean 0.61 (n=16, range 0.39–0.68, sd 0.10) against GLA's 0.40 (n=16, range 0.39–0.40); in campaign m, 0.46 (n=16, range 0.29–0.65, sd 0.12) against 0.40 (0.39–0.40) — and efficiency-adjusted composite 0.85 (range 0.50–0.94) against 0.51 (0.50–0.51) in q. The candidate's wide ranges are the run-level bimodality of §4.4 showing through on a second endpoint, and these deltas are descriptive, not gated claims; the per-run rows they summarize are in the released ledger (campaign m and q result files). We observed no degradation on the other tasks of the frozen suite in these campaigns.

5 Limitations

(1) **Scale, task, and what "natural language" means here.** All results are at ~14M parameters on synthetic MQAR-family tasks; this remains the top limitation, and the literature's base rate for such effects surviving real-corpus scale is poor. No claim here extends beyond S0. (Updated 2026-08-18: the p2g conversion certification (§4.3) extends the *predictive* claim to one 0.6B converted checkpoint, bounded by limitations 14–15; the mechanism and threshold claims remain S0-only.) Scale validation is carried by the law-transfer experiment (p1a/p1b: retrofit adapters on a pretrained model over a natural-language corpus, W ∈ {128, 256, 512, 1024}), which replaced an originally-planned four-architecture comparison run. Two caveats attach to it. First, a model trained on

ordinary web text alone scored 0% on the long-distance retrieval probe while copying nearby text at 87%, so the retrieval *format* must be supervised explicitly — the answer token is appended so that standard next-token loss teaches it. The law is therefore evaluated on a retrieval task **injected into** natural language, not on retrieval as it arises naturally in a corpus; the stronger reading is not supported by these data. Second, its W=512 cell failed its pre-registered gate (limitation 5).

(2) **Mechanism claims are configuration-bounded.** The component matrix of §4.6/§4.7 uses n=6 per arm on one panel/budget cell; "necessary" and "immaterial" are claims about this configuration, sealed only where §4.5 says so.

(3) **Budget, panel and step ceilings — what "attention did not cross" means.** It means: not within 1.32×10^{17} FLOPs (4.8×10^{16} across the learning-rate sweep), across base learning rates 10^{-6}–10^{-2}, on the `pairs80/p80x384` panel family, in 182 runs. It does **not** mean attention cannot acquire the task under this harness, and as of campaigns aw/ax we know it does: on an easy 48-token panel the same transformer crosses in 2/6 seeds at 3×10^{16} FLOPs (§4.10). The null is therefore jointly bounded by compute, by panel, and — see (11) — by optimizer steps. On the hard panels the transformer's threshold remains unidentified above a sixteenfold separation across the learning-rate sweep (fortyfold at the single step-matched rate) and is presumably finite, as both GLA's single crossing at 4× and attention's easy-panel crossing at 10× suggest. The sealed FoX-Pro-vs-transformer claim campaign (aq) — n=16 pre-registered pairs, plus transformer arms at escalated budgets × learning rates and a token-matched transformer control — has now completed and passed (16/16 vs 0/16, Fisher $p = 1.7\times10^{-9}$; all seven robustness arms 0/6). One corner of the transformer's hyperparameter space was untested when this limitation was first written: base_lr below 10^{-3} at budgets above 3×10^{15}, since the 3×10^{-4} sweep of §4.8 ran only at 3×10^{15} and aq's escalated budgets used base_lr $\geq 10^{-3}$ — and the learning-rate argmax then sat at the lowest rate tested (3×10^{-4}), leaving that axis not formally bracketed, although the increments decayed ($0.064 \to 0.115 \to 0.126$ at 3×10^{15}) — a value now identified rather than extrapolated: 0.1260 is the Bayes-optimal no-retrieval accuracy on this panel (§4.1.1), so the sweep converges on the ceiling of not-retrieving rather than approaching the 0.8 threshold. Campaign as ran that corner to its own pre-registration — base_lr $\in \{10^{-4}$, 3×10^{-4}, $5\times10^{-4}\}$ at budgets to 4.8×10^{16}, sixteen times the candidate's crossing budget — and returned zero crossings across its ten cells; its meta-gate on bracketing fired exactly as designed (the argmax was again the lowest rate tested), recording a further downward learning-rate extension as required. Campaign au ran that extension and the rate turned over rather than continuing to improve (§4.8): the learning-rate axis is bracketed on both sides, and this corner is discharged rather than merely disclosed.

(4) **Compute-fairness convention.** Matched measured FLOPs (C2) is one defensible convention; token-matched training would advantage attention and may change the inversion's magnitude (the aq design includes a token-matched arm).

(5) **Windowed complexity and the window boundary.** The supported statement is O(T·W) computation with recall given by the §4.1 law, not unbounded linear-time retrieval. On the synthetic panels no multi-layer relay beyond the window emerged within budget, but the natural-language transfer run (p1b) failed its pre-registered gate for the W=512 cell by overshooting (0.594 observed vs 0.415 predicted), and its retrieval-distance histogram

places 26 of 128 probes at a distance of exactly 512 — one token outside the strict predicate. That failure stands as reported. Campaign ar attempted to settle the boundary directly and **its result does not support a conclusion about reach in either direction**; it is reported in full in §4.9, together with the SWAX-documented difficulty confound that explains it, and the synthetic boundary question is open there. Campaign at has now reported and did not settle the question (INSTRUMENT_SUSPECT; see §4.9 for the two reasons), and a repaired-instrument attempt (`instrument-repair-20260811-au`; §4.9) also failed. Provisioning should therefore size W by the strict convention and treat any extra reach as margin, which is conservative under every hypothesis including the null. All windowed arms in this programme carry the full decay-attention component set; windowed vanilla softmax attention is untested here — at these budgets it does not transition at all (§4.10) — and the only plain sliding-window evidence is the zero-shot deployed-checkpoint series (p2a–p2f), which certifies the cliff but not saturation.

(6) **Out-of-sample law status.** The four-cell agreement is retrodictive; the ap test is pre-registered and its final aggregator decision (48 runs) is now complete and fully consistent (all 8 cells within 0.005). The law is a ceiling for transitioning runs, not an expectation over runs; §4.1 reports the per-cell seed counts.

(7) **Numerics.** Runs are fp32 without AMP; bimodality's sensitivity to numerics [2607.05104] is mitigated but not eliminated by C6 reproduction checks.

(8) **One window comparison is below the instrument's resolution and is withdrawn.** The law-transfer campaigns assert recall is non-decreasing in W across $W \in \{128, 256, 512, 1024\}$. For the first two cells the predicted ceilings are 0.2154 and 0.2231, a gap of 0.0077, while the 128-question battery resolves $1/128 = 0.0078$ per question and the standard error of the difference at $p \approx 0.22$ is 0.052 — about seven times the effect. That comparison cannot be decided by this battery in either direction, and campaign p1b's ordering gate 'passed' it by exactly one question, which is luck rather than evidence. We therefore claim ordering only across resolvable separations (W=256→512→1024) and report W=128 vs W=256 as untested. A larger battery (~23,000 probes per cell for half-gap resolution) is pre-registered against retained checkpoints. (Updated 2026-08-20: the withdrawal has since been discharged by p1d's redesigned 2,048-probe battery — predicted gap 0.215 against resolution 0.00049, observed 0.2148; see the §5 addendum, item (8, updated). The pre-registered ~23,000-probe battery is thereby superseded as the remedy, not exercised.)

(9) **Evidence retention in campaign p1b.** p1b's cloud instances were deleted before its per-cell results were copied to the lab's local GPU host; raw artifacts survive for one of its four cells, and the other three cells' numbers exist only as transcriptions in its decision record. Those three are not cited as evidence here. p1b did not pass its own pre-registered gate in any case, and campaign p1c re-tests the identical pre-registration on a repaired instrument with collection verified before teardown.

(10) **The interior probe is linear.** Campaign av establishes that the answer is not *linearly* decodable from a failed transformer's residual stream. A non-linearly encoded representation would not be detected, so the supported claim is 'not linearly present at any layer', which is the standard probing caveat, and the result is specific to one configuration (8 pairs, p80x384, 6×10^{15} FLOPs, d=384). It does not speak to whether attention acquires the

circuit at higher budgets. The instrument itself is no longer uncalibrated within-architecture: campaign ay reads 0.71–0.82 with the same probe in transformers that solved the task (§4.10), so a null reading is evidence about the model rather than about the probe's fit to attention. What remains open is the linearity caveat proper — a non-linearly encoded circuit would still escape detection — and the fact that the crossing transformers do not reach the candidates' 1.000, which we report without explaining.

(11) **Optimizer-step starvation under matched FLOPs — the control has now been run, and the null survived it.** Matched measured FLOPs (C2) charges the quadratic read against a fixed budget, so on a long panel the transformer receives fewer optimizer steps per FLOP than a linear mixer does. Until campaign az this was a live alternative explanation for the entire hard-panel null: the largest step count any transformer in that corpus had received was 21,797 (`pairs80` at 4.8×10^{16}), whereas the transformer's observed transition on the 48-token panel begins between step 40,528 and 60,792 — every hard-panel run was halted at roughly a third to a half of the step count at which this architecture is known to acquire the circuit. Because grokking-type transitions are indexed by optimizer steps rather than by FLOPs, that was a real confound and we declined to claim we had separated them. az is the control this limitation specified, run to its own pre-registration: `pairs80`, base_lr 3×10^{-3}, 1.32×10^{17} FLOPs, **59,943 steps**, six fresh seeds (2214–2219), transformer arm only, no efficacy stopping, with 0/6 and $\geq$1/6 given their meanings before any data existed. It returned **0 of 6**, maximum recall 0.1201, verdict `NULL_SURVIVES_STEP_MATCH` (2026-08-01T13:42:32Z). The hard-panel null is therefore not an artifact of step starvation at the rate that works elsewhere, and this limitation is discharged rather than merely disclosed. Two things it does not establish. It does not identify the transformer's threshold on the hard panels, which remains above 1.32×10^{17} FLOPs and unidentified. And it tests one learning rate at this step count; the step-matched sweep is a single cell, not a grid, so a reader who believes the hard panel needs both more steps *and* a different rate is not refuted by it.

(12) **The two load-bearing crossing runs reproduce exactly, which is a narrower result than "they replicate."** The conclusions of §4.10's ladder rest on seeds 2211 and 2212. Their source hashes, eval salt, panel, budget and token counts are byte-identical to the non-crossing seeds, and seeds 2210 and 2211 ran sequentially in the same process on the same GPU, which rules out a shard, salt or code confound. Campaign ay re-ran both on the byte-identical eval battery: both crossed again at recall 1.000, and seed 2212's final loss returned bit-identical to ax's ($6.816462725378613\times10^{-8}$). We state what that does and does not establish. It establishes that the runs are deterministic given the seed and are therefore not transient artefacts, corrupted evaluations or one-off numerical accidents — the failure modes a re-run is usually there to catch. It does **not** establish robustness to numerical non-determinism, because at fp32 without AMP on fixed hardware this harness exhibits none to be robust to; a re-run cannot test a source of variation that is absent. Nor does it add evidence about seeds: the transformer's crossing rate on this panel at this budget rests on ax's 2 of 6 and on nothing else, and 2/6 is what should be quoted. Earlier drafts of this limitation claimed the runs were untested "under fp32 non-determinism"; that framing was wrong and is corrected here rather than quietly dropped.

Addendum, 2026-08-01 (second) — prior art, and a claim downgraded. An adversarial prior-art sweep was run against this paper's own window-provisioning claim, before

external submission rather than after. It found the claim narrower than stated, and §1 Finding 1 (v0.4's Finding 4), §2 and §4.1 are rewritten accordingly. Three findings. (i) **The functional form is anticipated.** Yang [arXiv 2606.02680], June 2026, publishes Acc $\approx \pi + (1\text{-}\pi)\cdot$a_uncovered with the bound $\pi + (1\text{-}\pi)/K$ — algebraically our f + (1-f)·p_chance — and runs a needle distance sweep. We do not claim the equation. What survives is that we measure f from a task's retrieval-distance distribution rather than deriving π from mask geometry, and that we use it predictively and pre-registered rather than diagnostically. (ii) **The phenomenon is anticipated.** Sliding-window models failing beyond their window is reported in at least four independent places from 2024 onward (§2). Any reading of §4.1 as discovering that is a misreading and we have removed the phrasing that invited it. (iii) **The window-boundary convention is a known engineering ambiguity, not a discovery.** Microsoft's own Phi-3 model-card discussion documents `sliding_window=2047` as an inclusive offset encoding 2048 tokens, while Mistral's reference implementation is exclusive; the HuggingFace convention changed three times across versions. §4.2's inclusive-boundary result therefore describes *our* implementation and must not be generalised — a companion campaign measuring released checkpoints found HuggingFace's mask exclusive at W-1 on the same models, i.e. the opposite convention. What the sweep did *not* find, after searching Google Patents, WIPO and FreePatentsOnline, is any third-party patent claiming recall prediction from window coverage, and no published work that controls retrieval distance independently of context length on released checkpoints. That is the defensible territory and it is methodological.

Addendum, 2026-08-01 (first) — campaign p1d. Items 1, 5, 6, 8 and 9 above are updated by it; item 13 is new. These are written as dated corrections rather than edits in place, per this document's convention, so that what was claimed and when remains legible.

(1, updated) Scale validation is no longer carried by p1a/p1b. It is carried by **p1c** (training) and **p1d** (evaluation): a 110M-parameter windowed model on natural text, four windows, with all six of p1d's pre-registered gates passing and calibration inside 0.005 (§4.2). Two caveats from the original item survive unchanged and one is added. The retrieval format must still be supervised explicitly, so the law is evaluated on retrieval *injected into* natural language and not on retrieval as it arises in a corpus; the stronger reading remains unsupported. p1c's own composite gate reported **GATE_NOT_MET** on its ordering term and that verdict stands unamended — p1d is a separate, later campaign with its own pre-registration, not a re-scoring of p1c's gate. And the new caveat is item 13.

(5, updated — the boundary question is settled on this instrument, and the strict convention was wrong) The original item reported the boundary as **open**, because campaign `ar` varied window width and thereby confounded reach with optimization difficulty, and it named the inverted design — hold the window fixed, sweep retrieval distance across the boundary — as the only thing that would separate them. p1d's Battery B is that design. Across three windows independently (128, 256, 512), retrieval is intact at distance exactly W (0.922, 1.000, 0.906) and collapses to chance at W+1 (0.016, 0.000, 0.031). The supported convention is $\mathbf{d} \leq \mathbf{W}$, inclusive. Two consequences. First, the provisioning guidance in the original item — "size W by the strict convention and treat any extra reach as margin" — was conservative in the right direction and remains safe, but the extra token is now measured rather than hypothetical. Second, **the depth-relay reading is refuted, not confirmed**: a stacked-sliding-window receptive field (reach $\leq$ window $\times$ depth) would extend recall well past the boundary, and recall stops dead one token past W in every cell. Internal notes in the p1b and p1c records that filed this as a receptive-field effect are wrong and are superseded by this entry. What remains genuinely open

is the *synthetic* boundary question, which `ar` left confounded and which p1d does not address: p1d is one architecture on natural text, and campaign `at`'s inverted synthetic design has not been run.

(6, updated) Out-of-sample law status now has a second, independent leg. The synthetic leg is campaign `ap` (8 cells, pre-registered, all within 0.005). The natural-language leg is p1d (4 cells, pre-registered, all within 0.005, at 8× the parameter count). The two legs share the functional form and share nothing else — different scale, different corpus, different architecture family, different chance rate (1/16 versus 1/65). What neither leg establishes is model-level variance; see item 13.

(8, updated — the withdrawn comparison is restored, by a redesigned battery rather than by reinterpretation) The original item withdrew the W=128 versus W=256 comparison because on p1b's battery the predicted gap was 0.0077 against a 1/128 = 0.0078 resolution, and noted that p1b's ordering gate "passed" it by one question, which is luck. p1d's battery is a redesign, not a rerun: 2,048 probes stratified over 32 distances from 24 to 1,000, giving a resolution of 1/2048 = 0.00049 and a predicted W=128→W=256 gap of 0.215. The observed gap is 0.2148, roughly 440 times the resolution. The comparison is now decidable and it passes. The original withdrawal was correct on the evidence available then and is not retracted; it is discharged by a better instrument.

(9, updated) The evidence-retention gap that lost three of p1b's four cells is closed in practice, not only in policy. p1c's teardown was gated on sha256-verified collection on hub before any DELETE was issued, and its checkpoints survived to be re-scored by p1d — which is the specific thing p1b could not do. One defect in that teardown is corrected here: it recorded the first post-DELETE response, which was HTTP 200 (IBM's transient "deleting" state), and that ambiguous entry was later read as a possible billing leak. Both instance IDs were re-verified as HTTP 404 with zero instances and zero volumes remaining in the region on 2026-08-01. Nothing leaked. The confirmation step now polls to 404 rather than recording the first answer.

(13, updated 2026-08-04 — the natural-language law rested on one model; the pre-registered replication has now been run, and it passed) The original item stood as the top open weakness of the natural-language result: every number in §4.2 came from checkpoints trained at a single seed, 2100, so model-level variance was unestimated, and a single model can satisfy a ceiling prediction for reasons idiosyncratic to it. Campaign **p1e** was the test this limitation specified, run to its own pre-registration: seeds 2101 and 2102 across the same four windows, training recipe unchanged, a cross-seed gate requiring the per-window spread across the three seeds to remain within 0.10 against predicted between-window gaps of 0.21–0.27, with the readings — including the failure branches — fixed before any data and no efficacy stopping. It completed 2026-08-04T01:37Z with verdict **LAW_REPLICATES_ACROSS_SEEDS**: all seven gates (G0–G6) pass, seed 2100's four values reproduce exactly under the pre-registered instrument-reproduction check (agreement within 10^{-9}), all eight new cells calibrate within 0.005859 of their sealed ceilings against the ±0.05 tolerance, and the observed per-window spreads are 0.005371, 0.002930, 0.003418 and 0.000977 — more than an order of magnitude inside the 0.10 gate and roughly forty times smaller than the smallest between-window gap. This limitation is therefore discharged rather than merely disclosed, and per the reading fixed in advance the window-provisioning result may now be stated as a property of the architecture-and-window rather than of one model (§4.2). Two things p1e does not establish. n is 3, not large: the spread is an estimate from three models per window, and no variance claim beyond "small compared with the effect" is made from it. And it is one architecture, one corpus, one training recipe — the deployed-checkpoint transfer question of §4.3

and limitation 14 is untouched by it.

(14, NEW — deployed-checkpoint transfer is not claimed) The p2 series (§4.3) confirms the four-way dissociation and the boundary-convention observation on released checkpoints, and nothing more. The strict pre-registered transfer gates failed in both arms of both p2b and p2c, p2b's pre-registered LAW_DOES_NOT_TRANSFER outcome stands, and p2c's passing calibration is fitted rather than predicted — instrument characterization, not a result. The far-field residual (≈0.024–0.030 mean over far strata, maximum 0.083, against measured length-matched chance of 0.000–0.003) is real on the corrected instrument and unmodelled by the law. Its direction — deployed checkpoints recall slightly *more* than the fitted ceiling, and nothing is lost inside the window — means the law under-promises there, so provisioning with it errs conservative; a conservative instrument is still an uncertified one. The pre-registered p2d (instrument frozen as p2c leaves it, predictions fixed from p2c's fitted boundary and residual term, fresh probe draws) has now run: all three windowed arms passed their frozen per-stratum predictions, including a boundary the fit had never seen, but its window-disabled control arm declined at extreme absolute distance, and the follow-up p2e showed the same decline in a second checkpoint with a different RoPE base — so the pre-registered confound branch triggered and the passing arms remain suspended. The from-scratch diagnostic (p2f) completed as this draft was frozen: its own pre-registered positive-control gate failed — the W=4096 control arm produced no sharp step at this scale — so p2f is INSTRUMENT_SUSPECT under its precedence rules, no claim is drawn from it, and the suspension stands. No sentence in this paper claims the law transfers to deployed checkpoints. The p2g conversion pilot (§4.3) does not alter this: its converted model is trained with retrieval supervision in the corpus, which is the §4.2 regime, not zero-shot transfer.

(15, NEW 2026-08-18 — the conversion pilot certifies prediction, not product, and its record has stated gaps) The p2g pilot ran one training seed (2105); the registered design adds a second seed only after a certification pass, with a Llama-3.2-1B substrate second by signed order, and that follow-on has not run, so the conversion result is single-seed and single-substrate. Its random-placement control was skipped under a pre-registered degeneracy rule (all 28 layers converted leaves no non-trivial placement), so KILL(i) is not evaluable and is not reported as passed. The conversion failed its quality kill-gate by 3.4× the budget (+0.343 nats held-out at sequence length 2048; +0.356 at 8,192, reporting-only), against a comparison biased in the converted model's favour — it alone was fine-tuned on the corpus head — so the deficit is if anything understated; it attributes to the conversion — recipe and/or windowed architecture, not separated by this design — and not to the law, whose claim is recall only; the descriptive per-arm deltas (§4.3) locate most of the penalty in the shared conversion-training recipe. A registered descriptive re-window ladder off the A1 checkpoint was not executed; its registered skip clause (a cost cap) was never invoked, so the omission is a deviation rather than an exercised branch — the checkpoint is retained and the item remains executable. Two custody notes on the retained campaign record (available on request, §Reproducibility): the gated collector's manifest records `score: null` for every arm, because the key it reads to attest score collection was never written by the finalization script — the verdict's score files are instead verified against safety copies hashed at source on the instances before teardown, sound as evidence but lacking the collector's own attestation, which cannot be recreated since the instances are destroyed; and the finalized per-arm results record superseded failed attempts *inside* the artifact — A1's OOM-killed stage-2 attempt sits under `stage2_superseded_oom` beside a top-level `status: complete`, while an earlier safety-copy `result_A1.json` (2026-08-09) still carries the raw `failed_stage2`

status it was written with — so readers tallying attempts should read those keys and the stage run-records rather than a status field from either copy alone. The manifest gap is disclosed rather than repaired, because the underlying artifacts are sealed. A third custody note: the kill-gate's base-side holdout loss was first measured on a spot instance whose copy was destroyed at teardown, and was re-measured locally on 2026-08-13 (same pinned revision, corpus and window count; the superseding artifact records both facts) — so the +0.343 comparison pairs an instance-scored converted model against a locally re-measured base, on different GPU hardware; the converted-arm losses in the finalized results match the instance-scored values exactly, and the gate fails at 3.4× the budget on any reading of that pairing.

6 Conclusion

At small scale under a sealed matched-compute protocol, a windowed decay-attention layer saturates its window: attained recall lands on the ceiling f + (1-f)·p_chance, which retrodicts four prior cells to within 0.005, matched all eight cells of a pre-registered out-of-sample test to within 0.005 with predictions sealed before any training result existed, and calibrates to within 0.005 on natural text at roughly eight times the parameter count, replicated across three seeds — every replication cell inside 0.006, per-window spread under 0.006, where recall is a step function at the window, the boundary inclusive in our implementation (released checkpoints differ; §4.3). The instrument behind those tests — retrieval distance held fixed independently of context length — carries to released checkpoints, where it finds a sharp recall cliff exactly where the architecture enforces a window and none where it does not, though the strict transfer gates fail and transfer is not claimed. Where the law is instead used to *provision* a deployed checkpoint — converting it to the windowed module and predicting its recall before the conversion trained — the prediction lands 0.0031 and 0.0061 away at two windows, on a conversion whose LM quality failed its registered budget: the prediction certifies, the product does not (§4.3, limitation 15). The ceiling is conditional: recall is acquired by phase transition, whether a run transitions is governed by an architecture-dependent compute threshold — crossed at a tenfold-lower budget on an easy panel, and a more-than-sixteenfold-lower tested budget on the hard panels, by architectures combining a softmax read, input-dependent decay, and local KV preprocessing — and the component that decides it, FoX-Pro's data-dependent KV-shift or an equivalent short convolution, is one that perplexity scores as marginal. If this structure survives scale, window provisioning becomes an engineering calculation performed before training, and "which abilities can this budget buy" becomes an architectural question, not merely a size question. The experiments were run under a protocol — pre-registration, sealed evaluation, adversarial self-audit, and the public correction of our own v0.3 attribution error within one day of its refutation — that we offer as a template for machine-speed science that remains accountable.

Appendix C: superseded and pre-assay-repair campaigns

The sealed campaign record released with this paper (appendix-c-campaign-record.md) reproduces per-campaign decision records verbatim from the ledger, beginning at campaign m. This section extends that record backward to the campaigns that precede it and forward to superseded intermediates within it; it will be merged into the record for submission. (The appendix is lettered C to match the released audit package's file naming; there are no appendices A–B in this document.)

C.1 The pre-assay-repair era (campaigns a–j)

The campaign graph of §3.6 begins at campaign k. Ten earlier campaigns (a–j, 2026-07-22 to 2026-07-26) predate the frozen suite, the transition-count protocol of §3.4 in its final form, and the matched measured-FLOP budget accounting; a reader who pulls the released ledger will find them, so the paper enumerates them first. Two rules govern their standing: nothing from this era is cited as evidence anywhere above, and nothing from it was deleted — every decision JSON, including the failures, is in the released ledger, backfilled at the ledger repository's creation like all pre-2026-07-30 campaigns (see Reproducibility). The era's assay scored *mean recall* rather than transition counts; the bimodality finding (§4.4) is what later made mean-based assays untenable, and several of this era's difficulties are that fact announcing itself.

- **a (calibration-20260722-a).** First assay calibration, four panels. All four returned `valid: false` — champion, transformer and GLA controls were statistically inseparable on the era's `recall_long` metric, all arms at 0.06–0.08, i.e. chance — status `ASSAY_REPAIR_REQUIRED`. The assay, not any model, was the finding.
- **b (assay-repair-20260722-b).** The repair: four re-parameterized panels (p8–p32); selected p32, the one panel where the arms separate (GLA 0.95, hybrid 0.42, transformer 0.08). Status CALIBRATED.
- **c (gla-confirm-20260722-c).** Sealed paired hybrid-vs-GLA comparison on p32, 8 pairs (seeds 100–107): the GLA control outperformed the hybrid champion (mean Δ 0.36 in GLA's favour, sign-flip $p = 0.016$); status NO_BREAKTHROUGH. Reported for completeness; the arrow of this comparison is orthogonal to every claim above.
- **d (gla-generalize-20260722-d).** Panel generalization of c across p28/p32/p36 (seeds 200–203); mixed deltas in both directions; inconclusive.
- **e (variance-diagnostic-20260723-e).** Claim-ineligible variance diagnostic (seeds 300–302) quantifying between-seed spread on the era's panels.
- **f (stablegla-assay-20260723-f).** First StableGLA assay calibration; selected p44, the one panel of four where StableGLA's mean (0.69) sat interior rather than saturated (p36 and p52 at ~1.0) or at chance (p60). Status CALIBRATED.
- **g (stablegla-confirm-20260723-g) — PASS, superseded.** Sealed paired confirmation on p44, 8 fresh pairs (seeds 500–507): StableGLA mean recall 0.877 against GLA 0.066, mean Δ 0.81, positive pairs 8/8, one-sided exact sign-flip $p = 0.0039$; all six pre-registered gates passed. This was the programme's first sealed StableGLA result. It is superseded, not cited: it predates the frozen suite, the transition-count protocol and the FLOP-matched budget accounting, and its comparison is subsumed by the nine sealed campaigns of §4.5, which re-establish it under the stricter design. One pair (seed 505, $\Delta = +0.023$, candidate at 0.086) already shows the bimodality later made explicit: that candidate sat at chance while the other seven transitioned.
- **h (stablegla-extrapolate-20260723-h) — claim-eligible FAIL, disclosed.** The sealed paired extrapolation of g to the harder p64_long regime at the same compute budget, 8 fresh pairs (seeds 510–517). It failed: mean Δ 0.014 against a pre-registered $\geq$0.10 gate, positive pairs 6/8 against a $\geq$7 gate, and candidate mean recall 0.077 against an interior-band gate of 0.20–0.90 — **both arms at chance**. The sign-flip p (0.016) technically cleared its own gate, which is exactly why the interior-band gate existed: a "significant" difference between 0.077 and 0.062 is not a capability. The recorded interpretation stands: the p44 confirmation did not extrapolate to p64_long at that budget. In this paper's

later vocabulary the diagnosis is panel/budget miscalibration — the regime sat below the candidate's compute threshold — a reading campaign i's diagnostic then supported directly. h is the programme's first claim-eligible negative result; it was recorded as FAIL inside its own campaign, and no claim above rests on the regime it tested.

- **i (stablegla-compute-threshold-20260725-i).** Claim-ineligible diagnostic that turned h's failure into the compute-threshold hypothesis: paired GLA/StableGLA arms at 1×10^{15} and 3×10^{15} FLOPs. StableGLA moves from chance (0.063) at 1×10^{15} to 0.998 at 3×10^{15} while GLA stays at chance at both; decision `COMPUTE_THRESHOLD_SUPPORTED`. This observation is what the compute-condition programme of §4.4–§4.5 grew from. A second ledger directory, stablegla-mechanism-20260725-i, shares the suffix and is empty — a registered-then-abandoned placeholder, disclosed here as a suffix collision alongside the seed collisions disclosed in §3.4.
- **j (stablegla-interior-calibration-20260726-j).** `PRELAUNCH_ABORTED`: a contract check expected mixer_api.py in each lane before it had been copied; no workers launched, no results, scientific change recorded as none.

The era's summary: two assay calibrations, one repair, one superseded PASS, one claim-eligible FAIL with both arms at chance, one aborted launch, and the diagnostic that produced the compute-threshold hypothesis the rest of this paper tests. The frozen-suite era (campaign k onward) rebuilt the assay around transition counts precisely because this era's mean-based scoring produced panels that were saturated, at chance, or unstable.

C.2 Superseded intermediates within the current era

Campaign p (stablegla-hardening-20260728-p) produced the write-gated positive-kernel "minimal mixer" of the first hardening round, which transitioned 3/6. It is a distinct architecture from the sealed minimal architecture of §4.6 and is retained here so the two are not conflated: the sealed minimal architecture (convolution + input-dependent decay bias + softmax read + QK-norm + output-norm) was identified by additive reconstruction and sealed in campaign ag; the hardening-round mixer is a superseded intermediate and no claim rests on it. Campaign g above is superseded in the same sense — a genuine sealed PASS whose comparison was re-established under the stricter frozen-suite design before anything was claimed from it.

Reproducibility

All campaign directories (manifests with pre-registered gates and source SHA-256 hashes, sealed salt hashes, per-run results with full training logs, decision JSONs, the operator's append-only decision log and seed ledger) are available at https://github.com/thebotclub/archlab-ledger (public). Every pooled transformer count in this paper — runs, distinct seeds, panel × budget × learning-rate cells, budget span and maximum observed recall — is recomputed from those artifacts by `recount_transformer_corpus.py`, released in the ledger repository; it is the single source of truth for those figures, and the hand-maintained counts of earlier drafts (88, 168, 172) are superseded by its output. The per-panel Bayes no-retrieval floors of §4.1.1 are likewise computed rather than asserted, by `mqar_floor.py`, likewise released in the ledger repository; it takes a panel's pair and query counts and the value-alphabet size and returns the floor by simulation, and we recommend running it before sealing any panel of this family. `recount_transformer_corpus.py` reports the hard-panel null corpus, the aw/ax falsification ladder and the early calibration panels separately, because they are separate claims. Campaign

manifests are additionally committed to an append-only ledger repository. The repository was created 2026-07-30T03:26Z — mid-way through campaign ap's training run: ap's sealed predictions enter at the inception commit itself (13 minutes before its first result artifact), and every campaign sealed after inception — aq onward, including the aw/ax falsification ladder, az, p1c's mid-training addendum, p1d, p1e and the p2 series — has its manifests committed contemporaneously by a 20-minute cron, whose commit timestamps establish pre-registration ordering externally. Campaigns sealed before that date (m through ao) were backfilled en masse at repository creation; their pre-registration ordering rests on internally consistent manifests, hashes and file mtimes, attested only by the machine that produced the results. Hardware: 4× NVIDIA V100-32GB for the harness campaigns; the p2g conversion arms trained and scored on 2× cloud L40S spot instances. Software: PyTorch 2.10, fp32, no AMP. The zero-shot deployed-checkpoint campaigns (p2a–p2e) are evaluation-only; their campaign manifests and decision JSONs are in the public ledger — including the void record of a name-collision campaign (prv-lawtransfer-20260802-p2b), retained rather than deleted — and their per-probe result files and run logs are retained in the program's Lab-2 run directories, available on request. p2f's control-arm result JSONs are in its ledger directory. For p2g the public ledger carries the sealed registration and the campaign manifest with battery hashes; the per-arm finalized result JSONs, stage run-records, and sha-verified safety copies of checkpoints and score files taken from the instances before teardown are retained by the authors and available on request; limitation 15 enumerates this record's custody gaps. One further note on provenance. This document carries a full dated changelog — including the v0.3→v0.4 refutations of our own claims and the v0.5 reframe — as banner entries that the arXiv build excludes by design. That changelog is preserved verbatim in the ledger repository, together with a full copy of this preprint, so what was claimed, when, and what refuted it remains reconstructible from the release even where the arXiv text does not carry it. Readers auditing the claim history should consult the repository copy.

Acknowledgments

The archlab autonomous operator — an instance of Claude (Anthropic) — designed and executed campaigns r, s, t and u end-to-end and executed the remaining campaigns under supervision, as described in §3.6. All claims were verified against primary artifacts.

Competing interests

The Bot Club Pty Ltd has filed Australian provisional patent applications 2026906694 (30 July 2026), 2026906768 (2 August 2026) and 2026907105 (18 August 2026) covering, respectively, the window-provisioning method described in §4.1; the distance-controlled measurement instrument and behavioural boundary-detection method described in §3; and the staged conversion-training and multi-gate certification methods described in §4.3. The author distributes recallkit, an open-source implementation of the provisioning method (Apache-2.0 with patent grant), and has a commercial interest in its adoption.

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